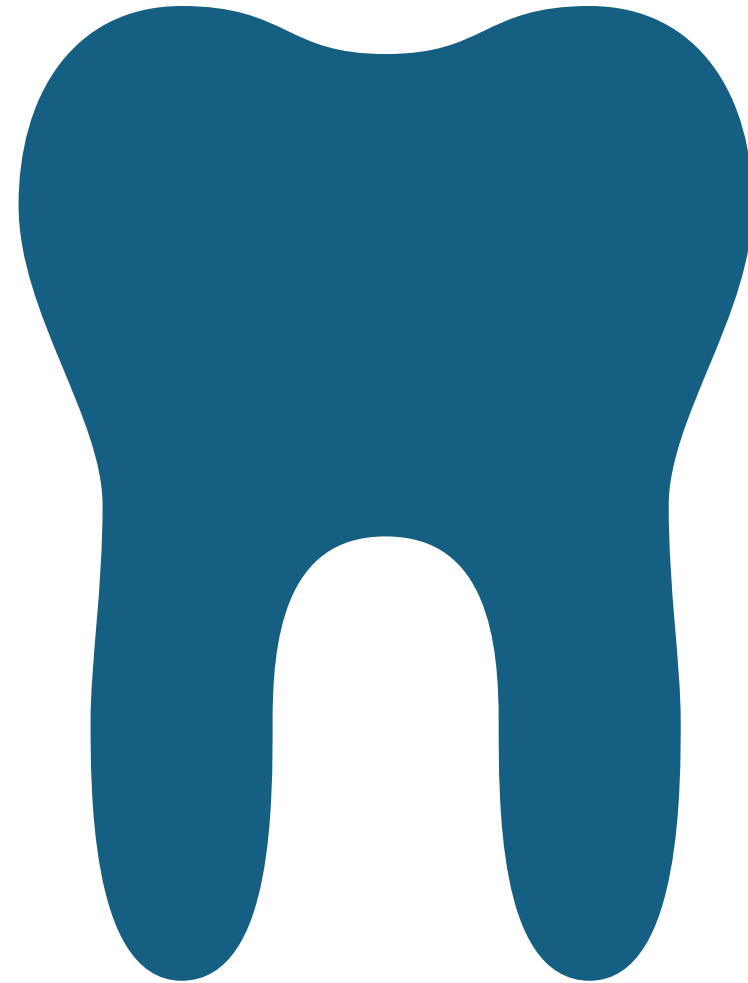


Pulp Chambers and Canals of Primary and Permanent Anterior Teeth

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Learning Outcomes

- **To Define** key pulp terminology and explain age/irritation effects on pulp size.
- **To Describe** pulp chamber, horns, and canals of each permanent tooth from all views; compare between tooth types.
- **To Recognize** common variations (multiple canals, MB2, apical delta) and age-related shape changes.
- **To Relate** pulp anatomy to clinical risks (exposure, fractures), mandibular canal proximity, and endodontic access.
- **To Compare** deciduous vs. permanent anterior pulp (larger, longer horns, two-canal incidence) and clinical implications.

Pulp Cavity: Terminology & Fundamentals

- ***Pulp chamber / Pulp cavity / Coronal pulp*** – refer to the coronal portion in crown of the tooth.
- ***Root pulp / Radicular pulp / Pulp canal / Root canal*** – refer to the radicular portion in root of the tooth.

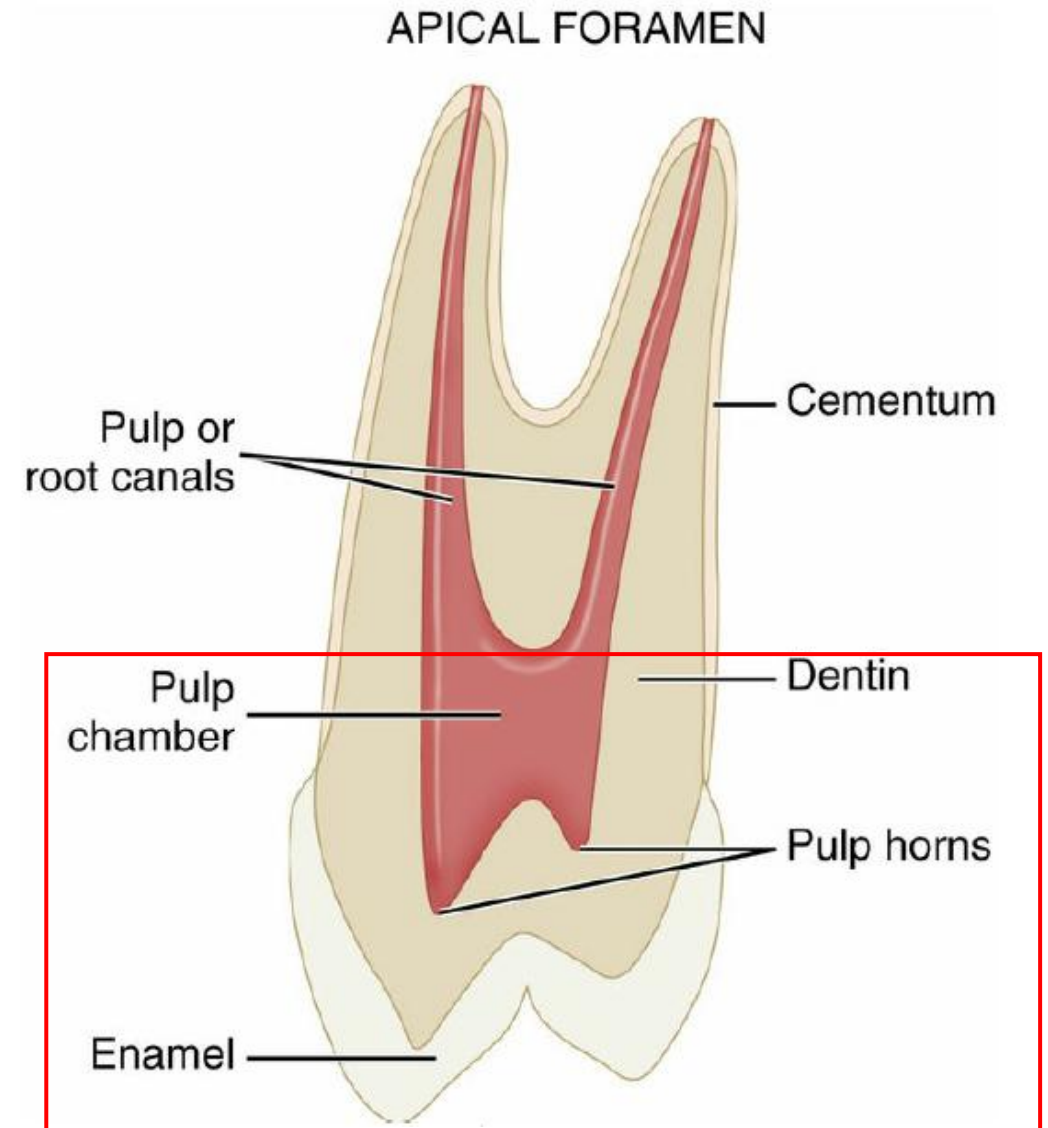


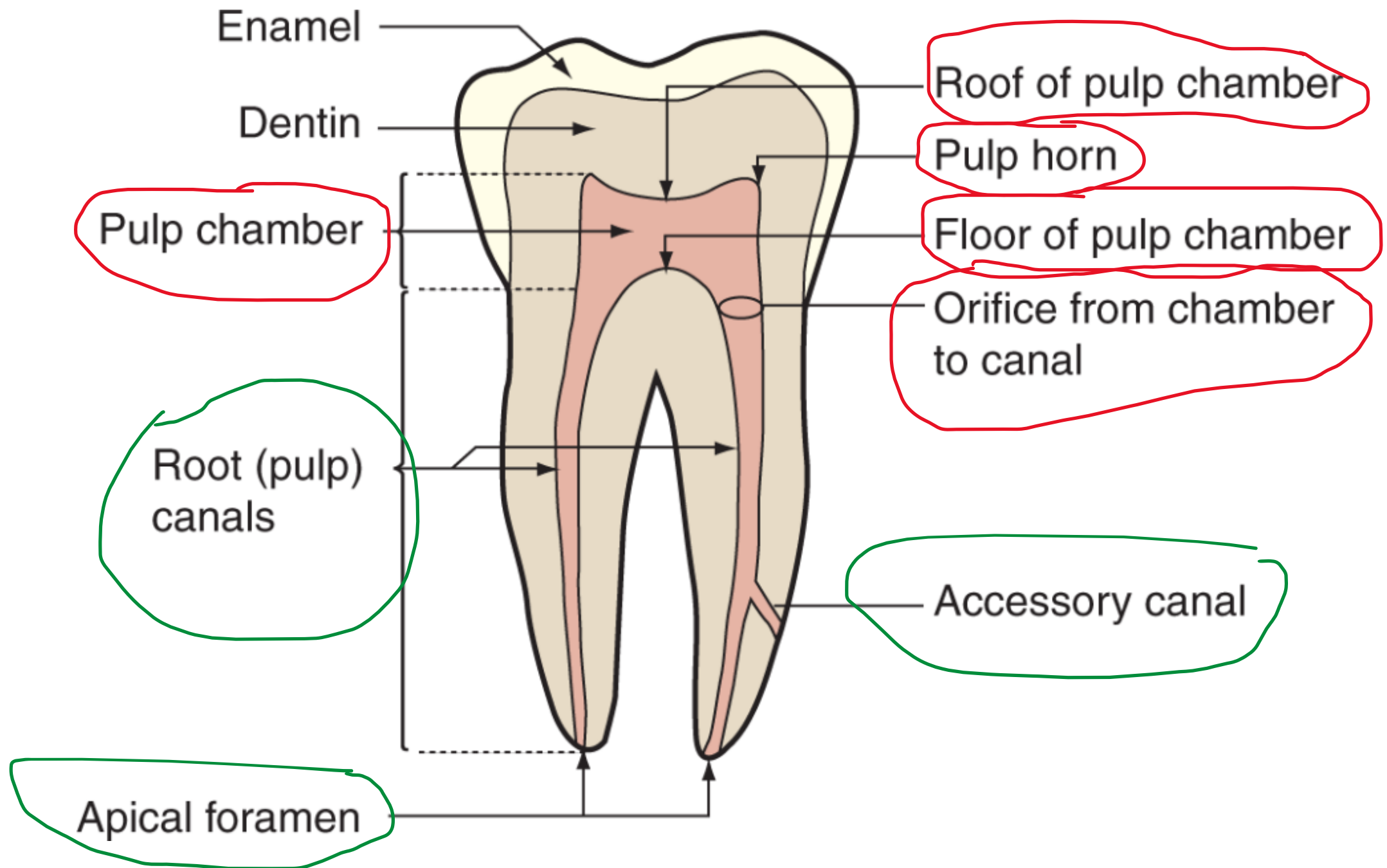
FIG. 13.1 Buccolingual section of a maxillary premolar.

Correspondence Rule:

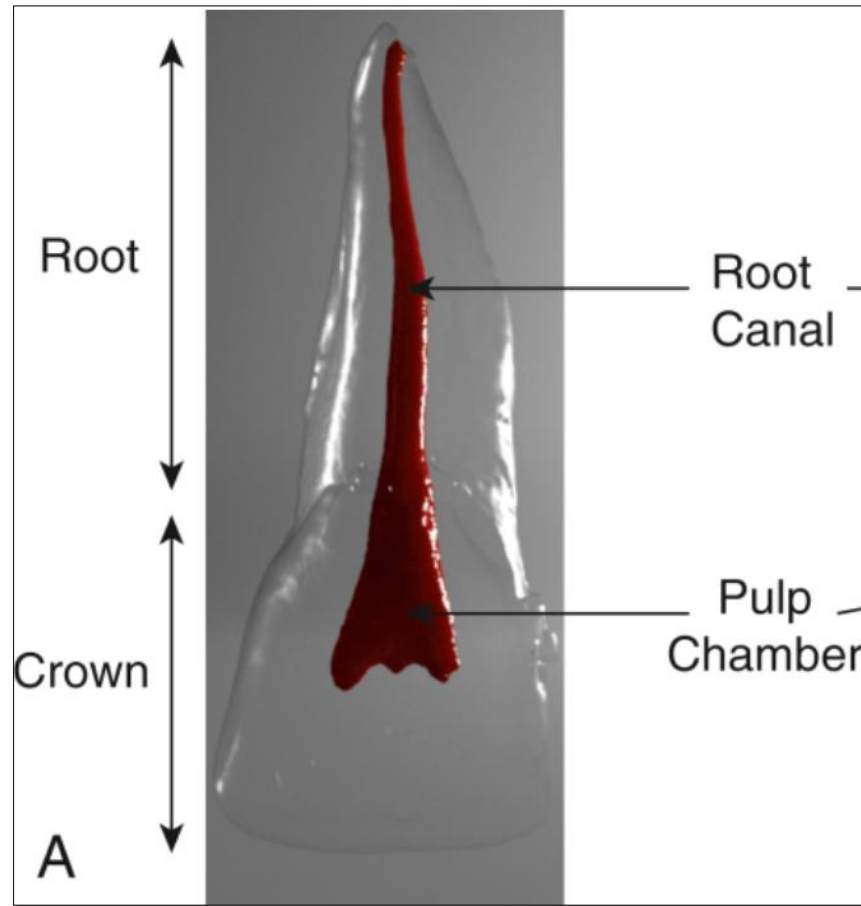
- Outline of pulp tissue generally follows the external outline of the tooth.
 - Pulp chamber shape $\leftrightarrow$ crown shape.
 - Pulp canal shape $\leftrightarrow$ root shape.



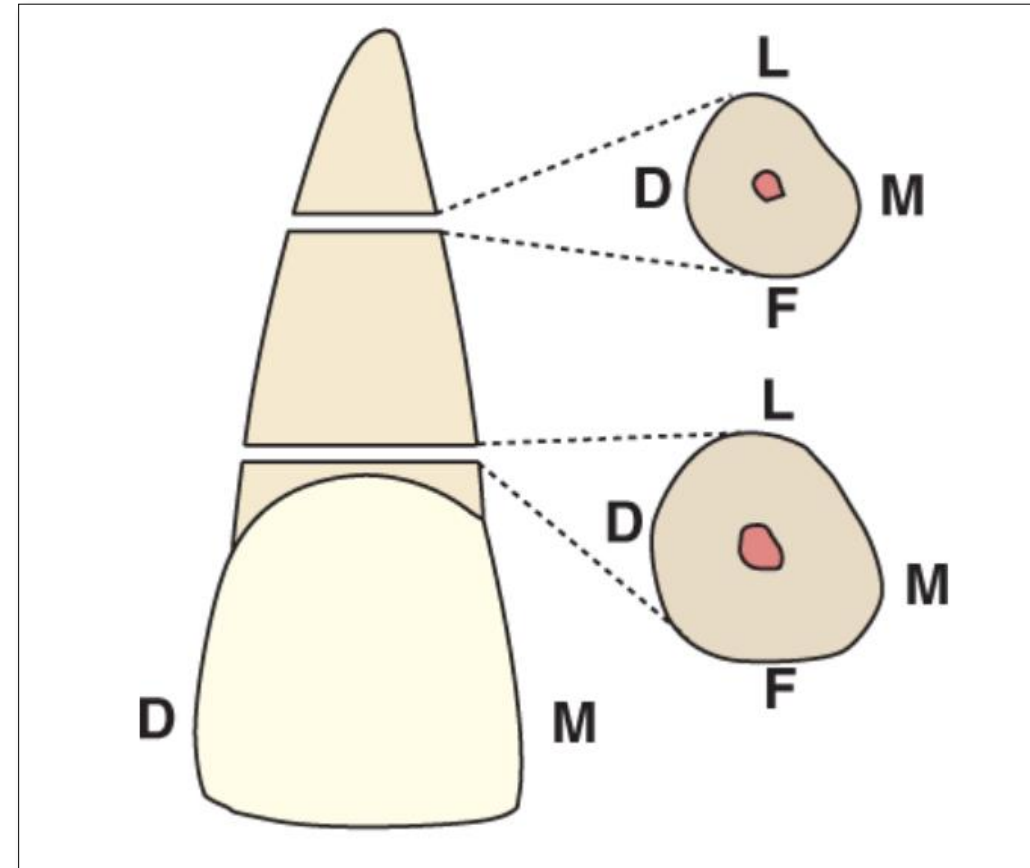
Maxillary Central Incisor



Anatomical Division:



Longitudinal Section

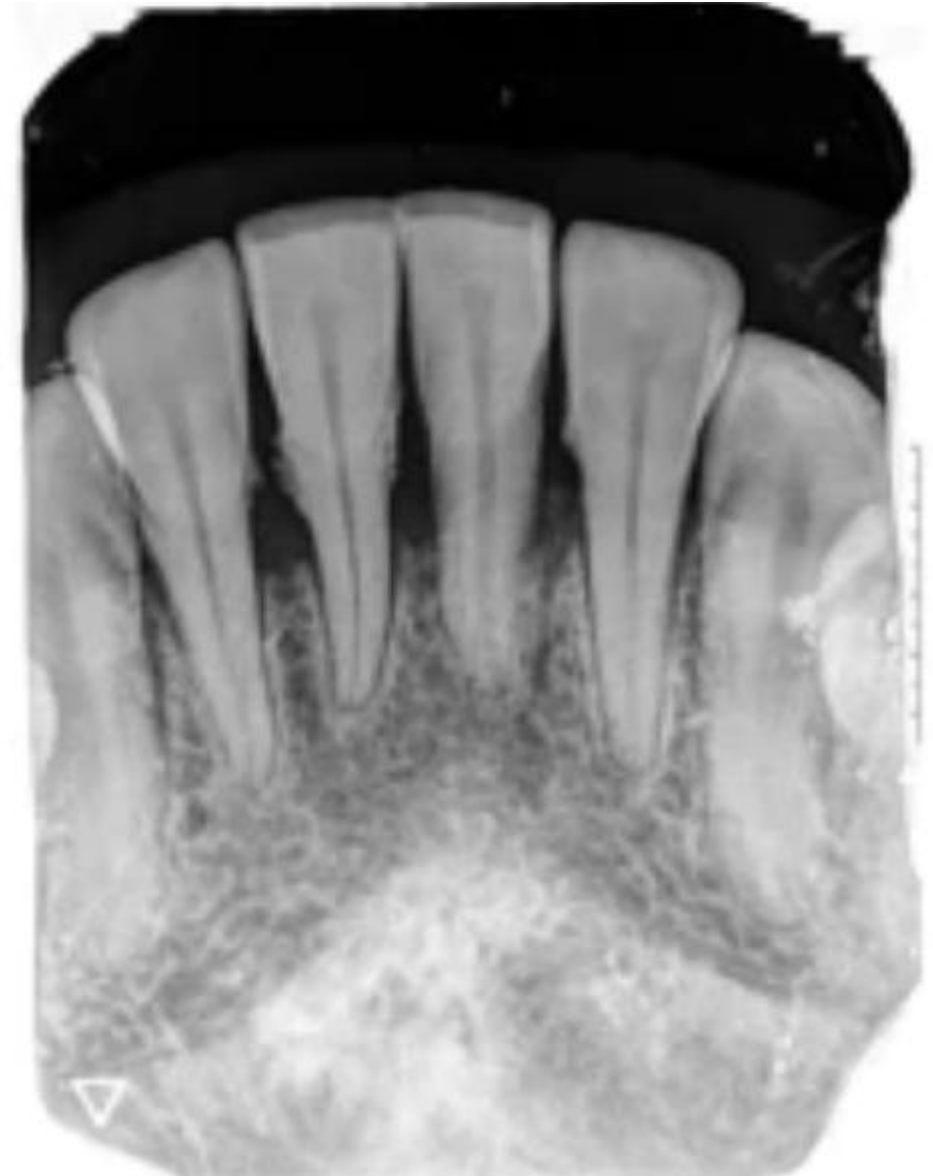


Transverse Section

Complexities best appreciated through **longitudinal and transverse sections** of tooth types.

Radiographic Visualization of Pulp Cavities

The pulp chamber and canals are 3D structures seen as 2D on radiographs. A **periapical (PA) radiograph** shows the **mesiodistal aspect** of entire anterior tooth from incisal edge to 2–3 mm beyond the apex along with roots, pulp chamber, and surrounding bone while the **buccolingual aspect** is often best visualized via cross-sectional imaging or specialized angulated X-rays.



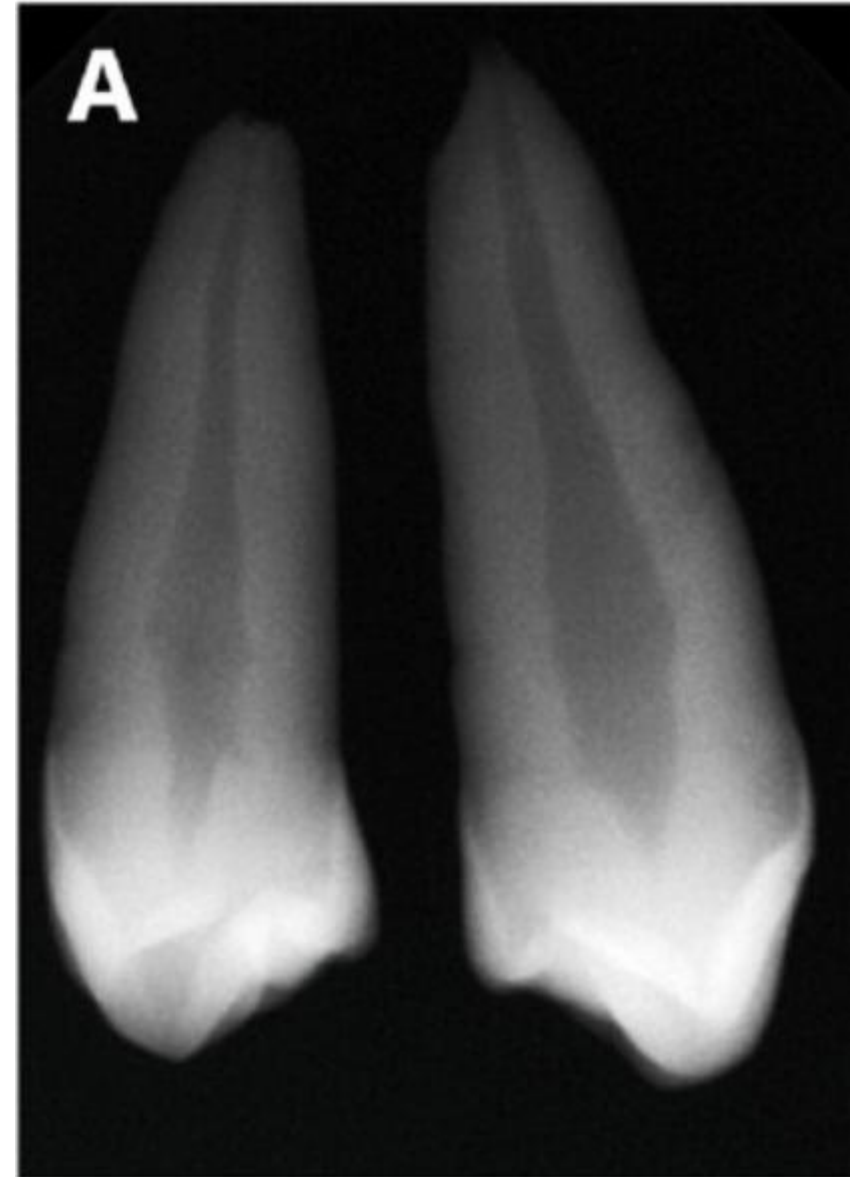
Standard View (Facial Orientation):

- Provides a **monoplane, buccolingual view**.
- Hard tissues (enamel, dentin) appear radiopaque (light).
- Pulp spaces appear **radiolucent (dark)**.



Mesiodistal Aspect:

- **Rarely seen** in routine **PA radiographs**.
- May appear incidentally in:
 - Malpositioned teeth
 - Rotated teeth
- Consequently, the **mesiodistal radiographic anatomy of pulp cavities is not well known**.
- Other ways are
 1. **Panoramic radiograph (orthopantomogram)**
 2. **Bite wing radiographs**



Size of the Pulp Cavity

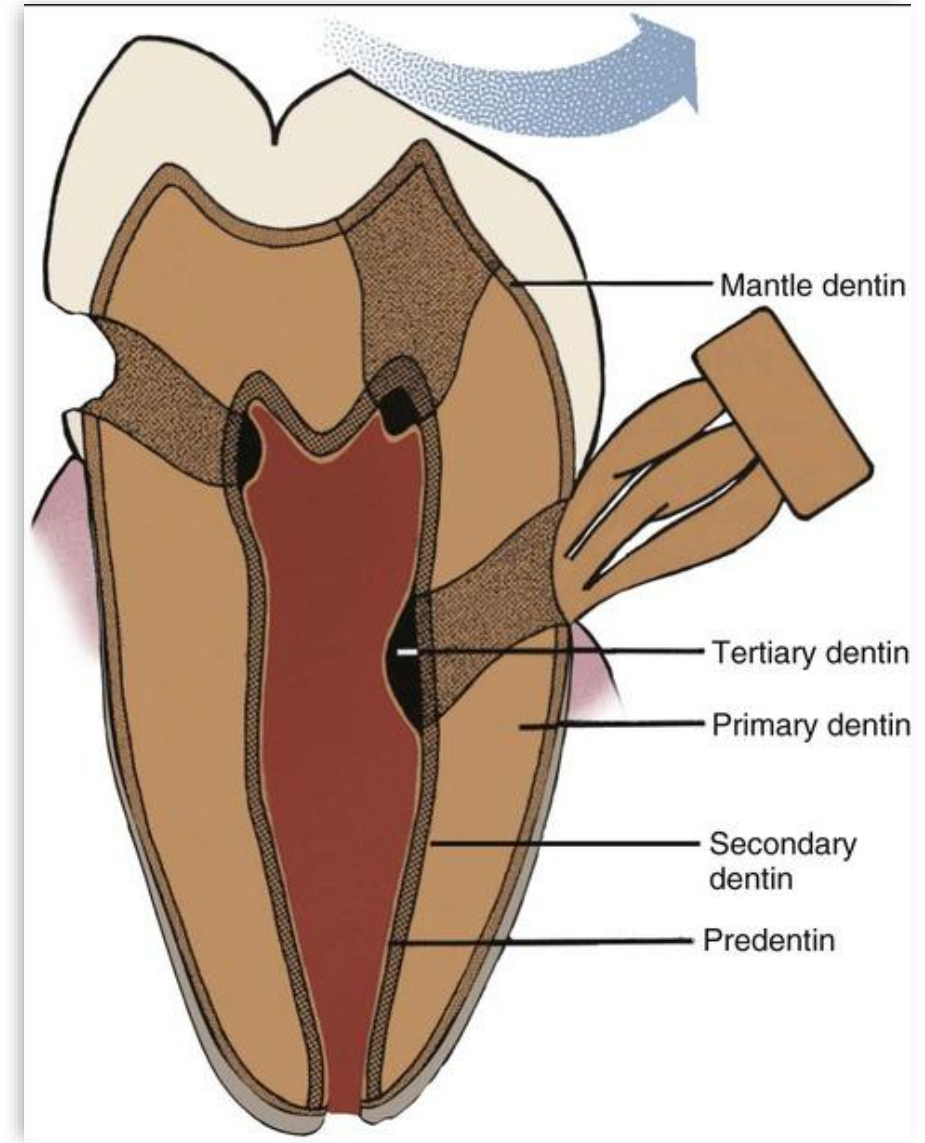
- **Key Factors:** Age and trauma history.

1. Secondary Dentin (Physiologic):

- Forms continuously throughout life.
- **Non-uniform:** More deposited on pulp chamber floor and roof.
- **Result:** Pulp cavity is **larger in young** than in adults.

2. Reparative/ Tertiary Dentin (Irritation-Induced):

- Triggered by caries, abrasion, attrition, operative procedures.
- **Protective but can eventually compromise pulp space.**



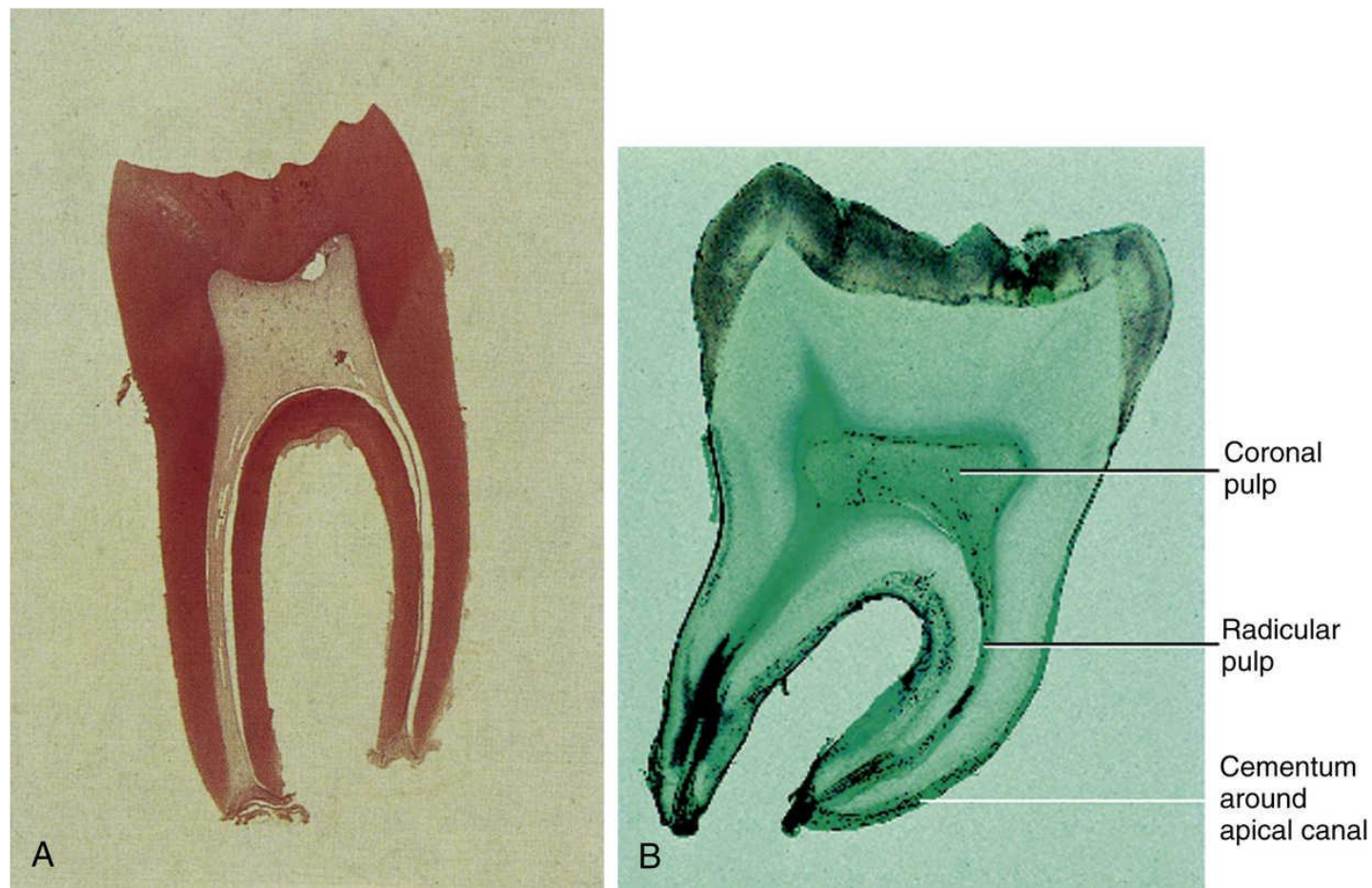


FIG. 13.3 Comparison in size of the pulps of two intact lower first permanent molars at different ages.

(A) The young pulp chamber is Large.

(B) The older pulp is greatly reduced in size

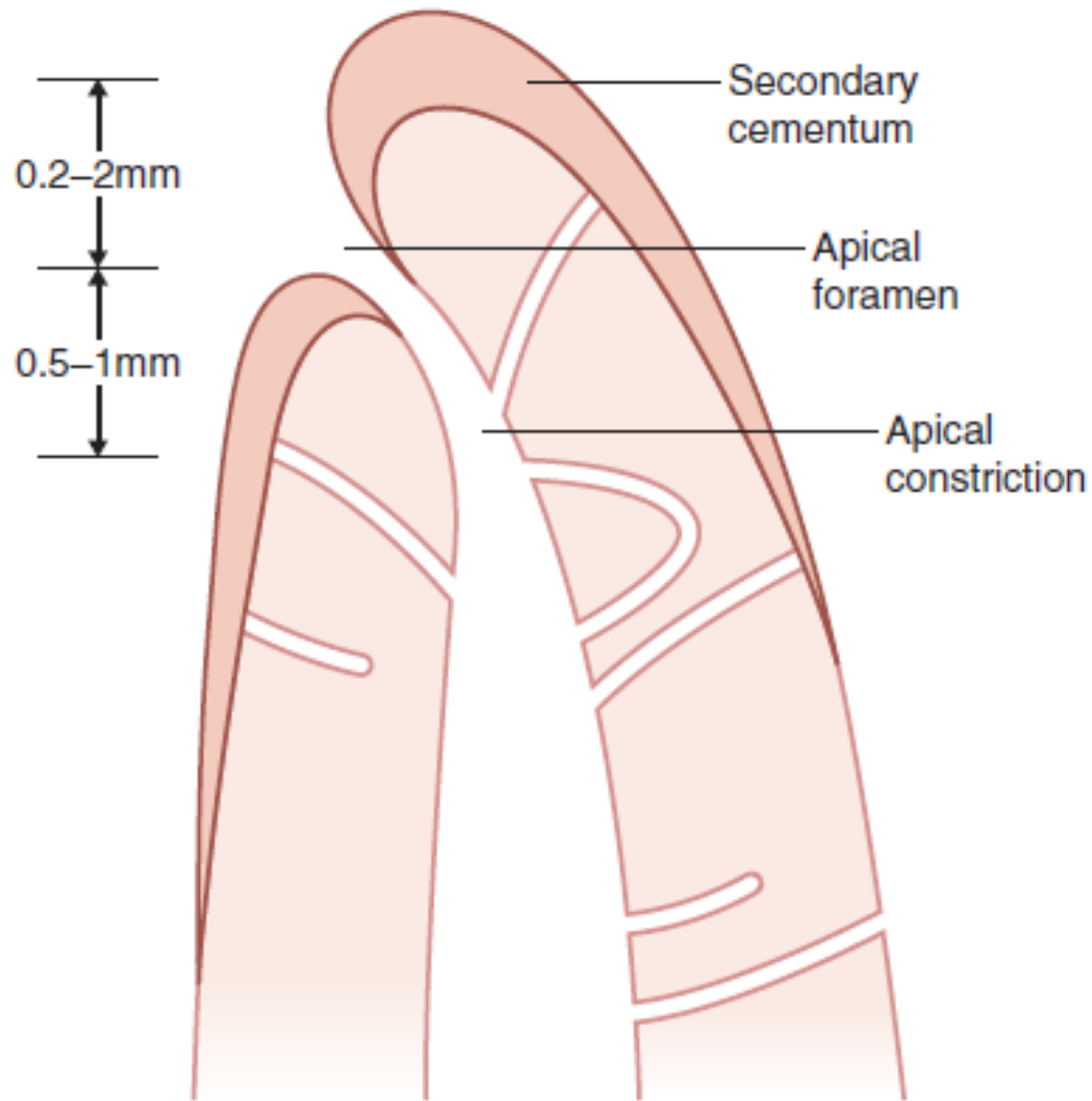


Fig. 2.59 Diagrammatic section through apical third of root showing position of apical constriction.

The Apical Foramen and Variations

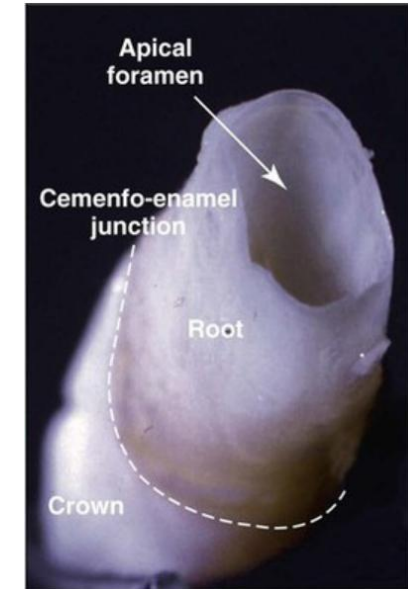
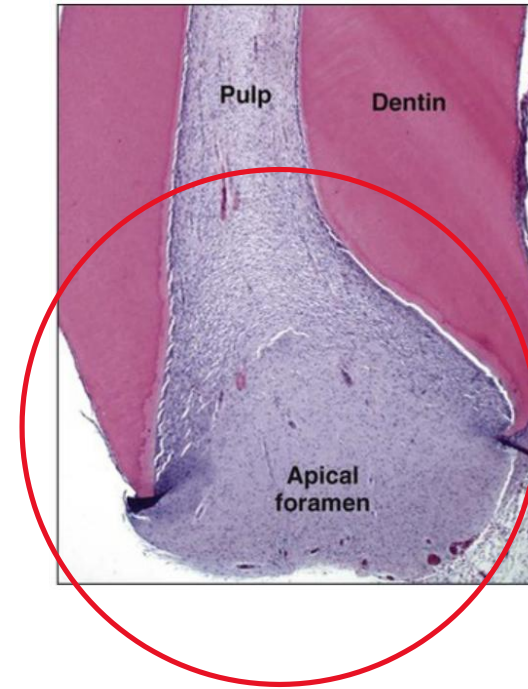


FIG. 13.4 Maxillary canine. Labiolingual sections show various stages of development.

- 1, Crown complete; root partially completed with large pulp cavity, wide open at apical end.
- 2, Tooth almost complete except for lack of constriction of apical foramen.
- 3, Canine of a young individual with large pulp cavity and completed root tip.
- 4, Typical canine of an adult, demonstrating constriction of the foramen.
- 5, Canine of an elderly individual with a constricted pulp chamber and canal

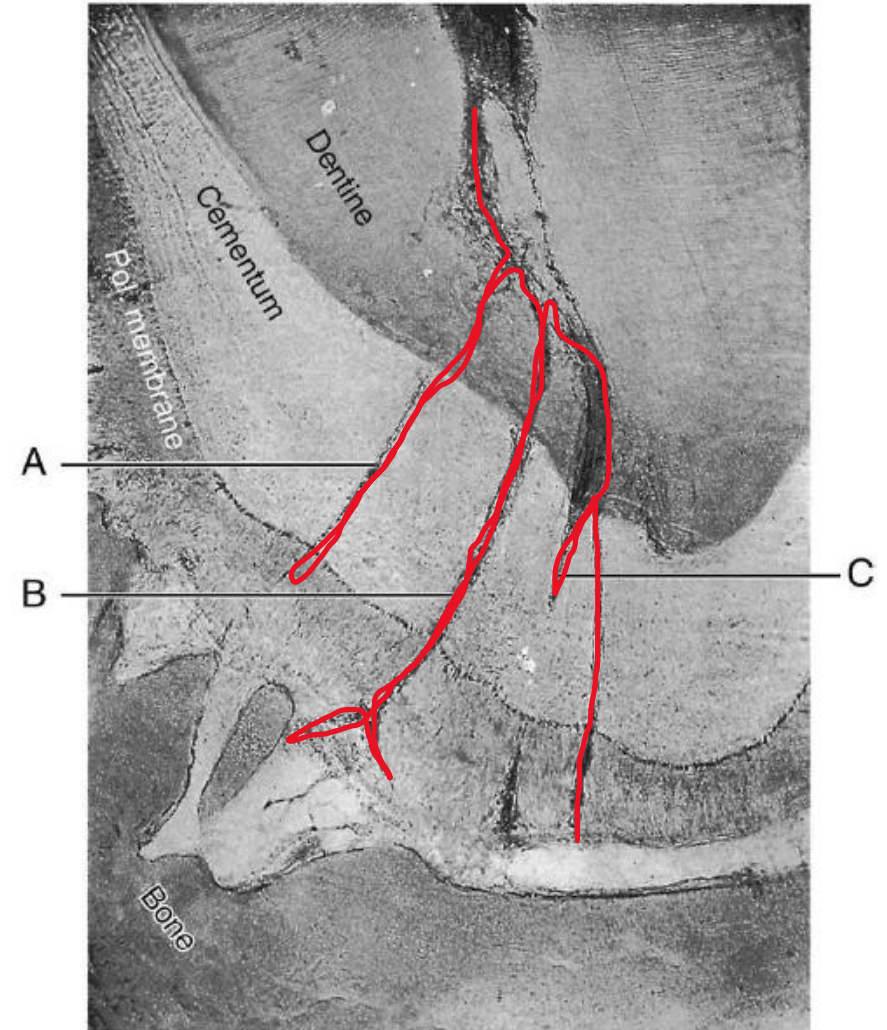
Size of the Pulp Cavity

- Compare pulp size with adjacent teeth.
- **Extensive localized calcification** → consider **elective endodontics** before complex restorations.
- Always account for patient age prior to extensive tooth reduction.

Variations in Root Apex Anatomy:

Accessory Canals/ Supplementary or Lateral canal: Small, isolated side branches off the main canal that exit on the root surface, usually in the middle or apical third.

FIG. 13.5 A section through the apex of a root showing multiple canals (A through C). Three canals are present within the dentin, but one canal divides at the cementum-dentin junction (C), which makes a total of four small canals that exit on the cemental surface of the tooth.

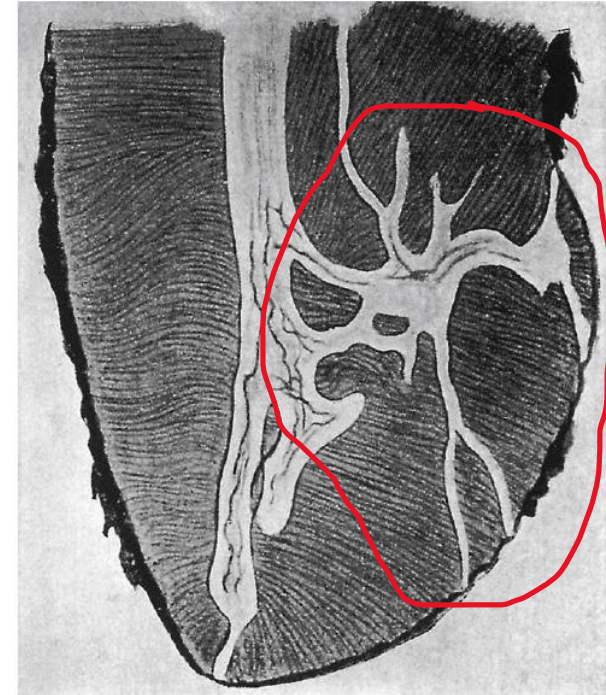
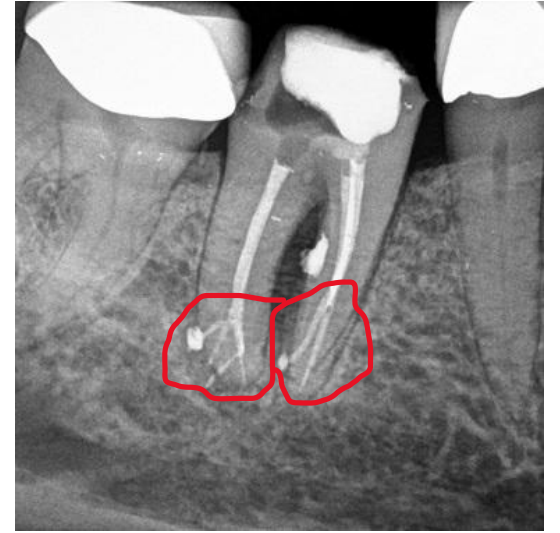


Variations in Root Apex Anatomy

Apical Delta: The main canal divides into multiple tiny canals near the apex, creating a complex network, near the apex, replacing the main canal.

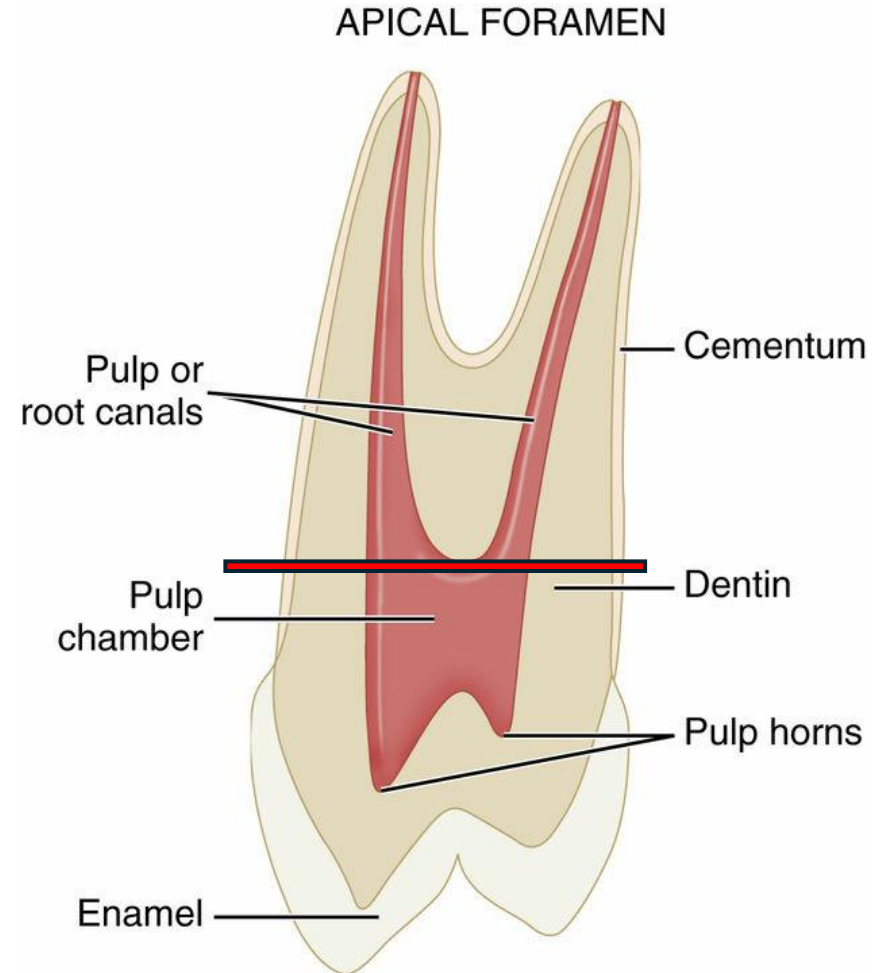
✓ Clinically difficult to instrument;
requires thorough irrigation.

FIG. 13.6 Apical end of root showing one main canal and an adjacent **delta system**.



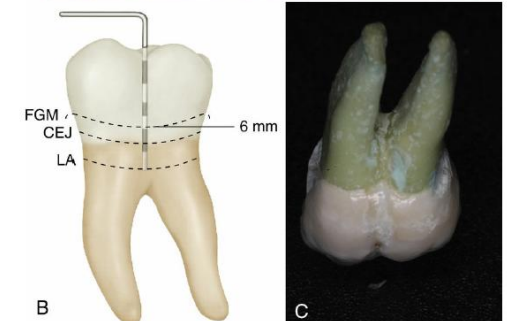
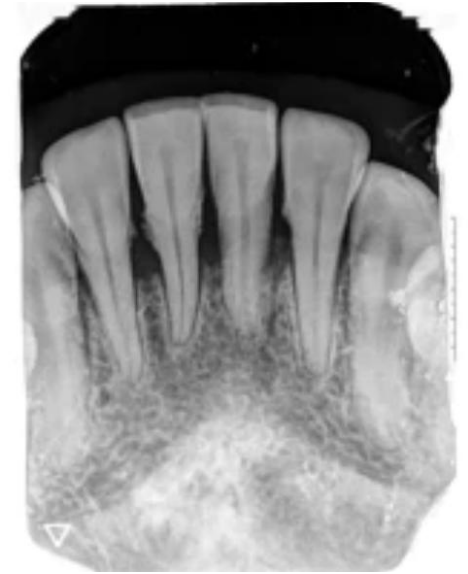
Demarcation of Pulp Chamber and Root Canal

- **Macroscopic Demarcation:**
 - **Cementoenamel junction (CEJ)** is a general landmark, but the transition occurs **at or near** this level.
 - In **multirooted teeth**, demarcation is simpler:
 - ✓ pulp cavity in root = root canal;
 - ✓ remainder = pulp chamber.



Macroscopic Demarcation:

- **Cementoenamel junction (CEJ)** differentiation aided by:
 1. Exploring CEJ
 2. Radiographic density difference:
 - ✓ **enamel** (crown) vs. **cementum** (root) covering the dentin.

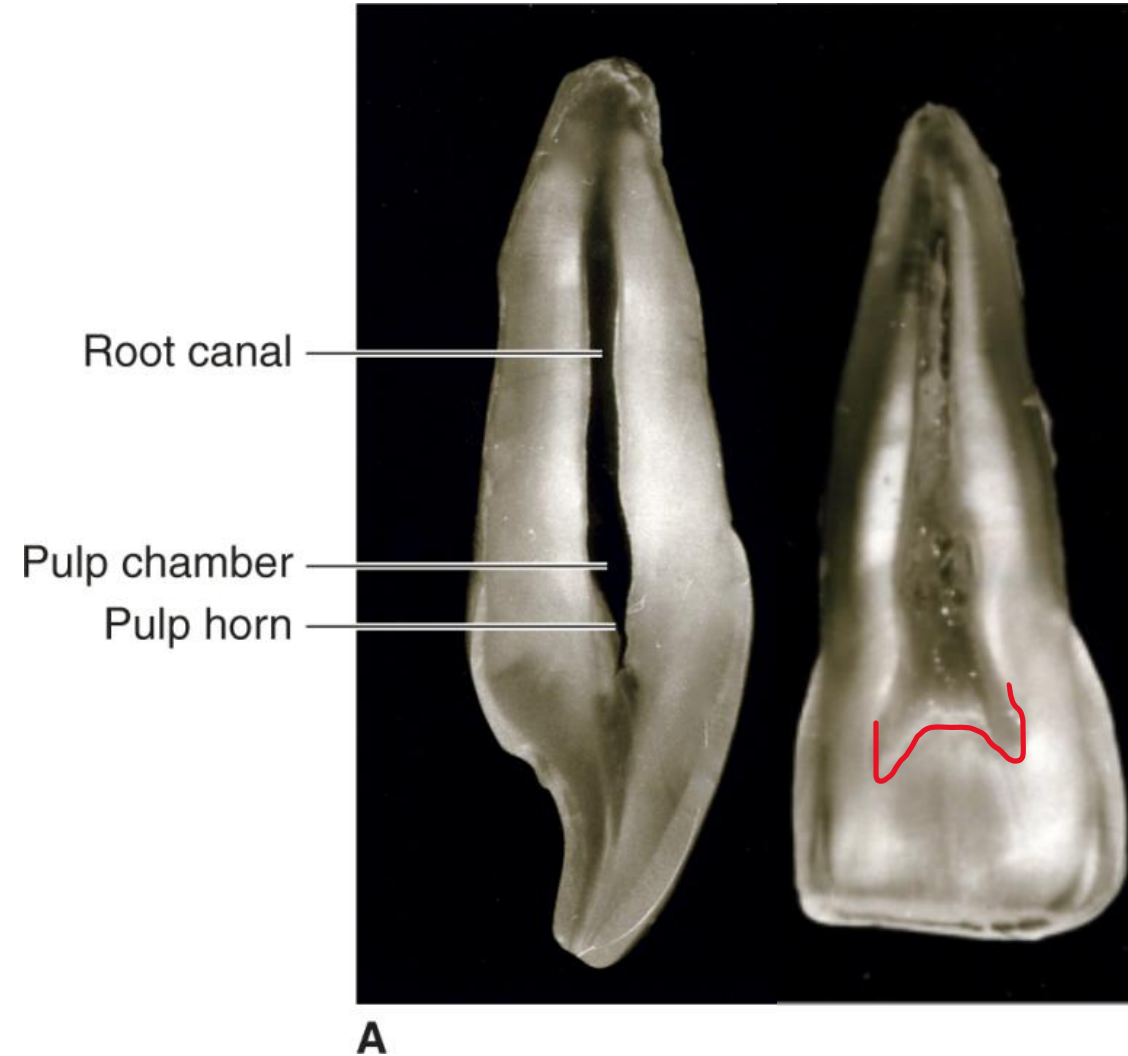


Microscopic Demarcation:

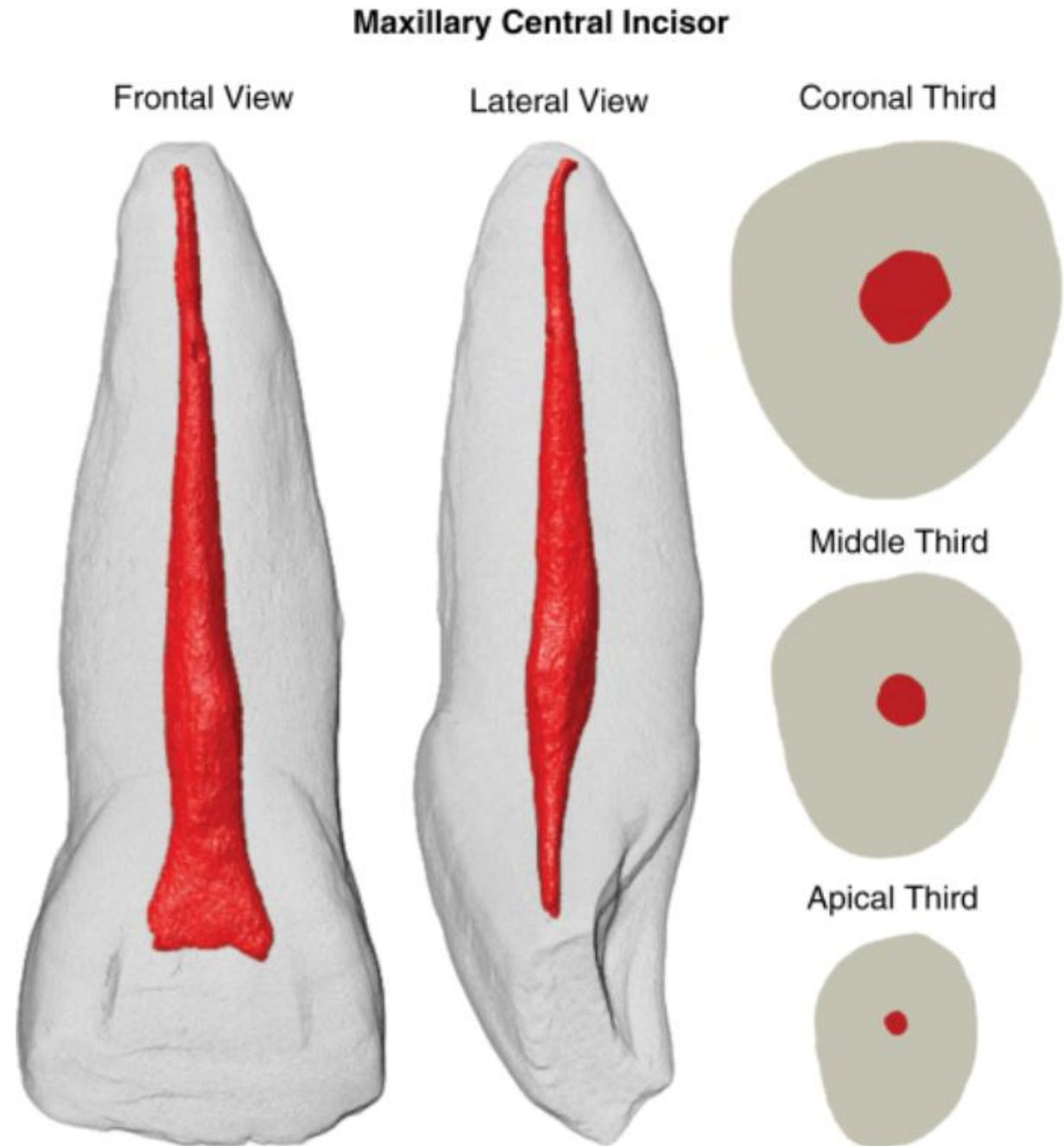
- **Pulp chamber:** More cellular; odontoblasts are **cuboidal**.
- **Root canal:** Less cellular; odontoblasts gradually **flatten** toward apex.
- Transition is **not sharply demarcated** microscopically.
- **Conclusion:**
 - The division between pulp chamber and root canal is a **macroscopic and clinical convenience**, not a precise histological boundary.

Pulp Horns

- **Definition:** Projections in the pulp chamber roof corresponding to crown cusps/lobes.
- **Correspondence:** Prominent cusps/labial lobes → prominent pulp horns (especially in youth).
- **With age:** Reduced prominence due to secondary dentin deposition.



Maxillary Central Incisor – Pulp Cavity



Maxillary Central Incisor – Pulp Cavity

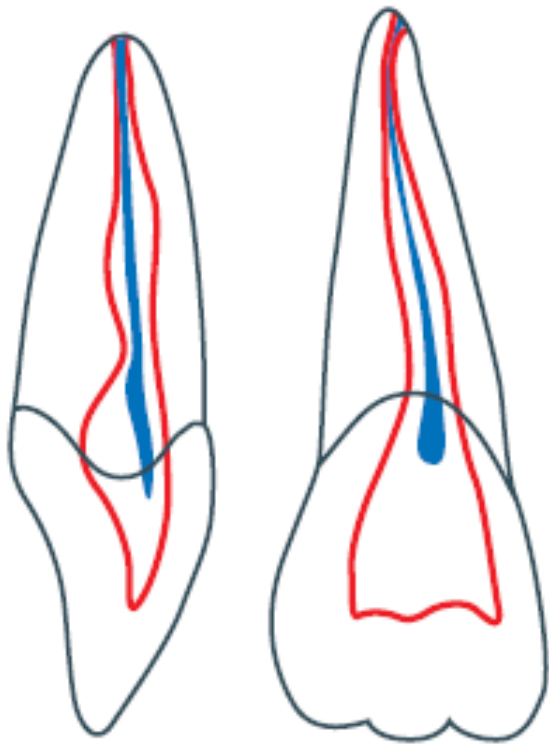
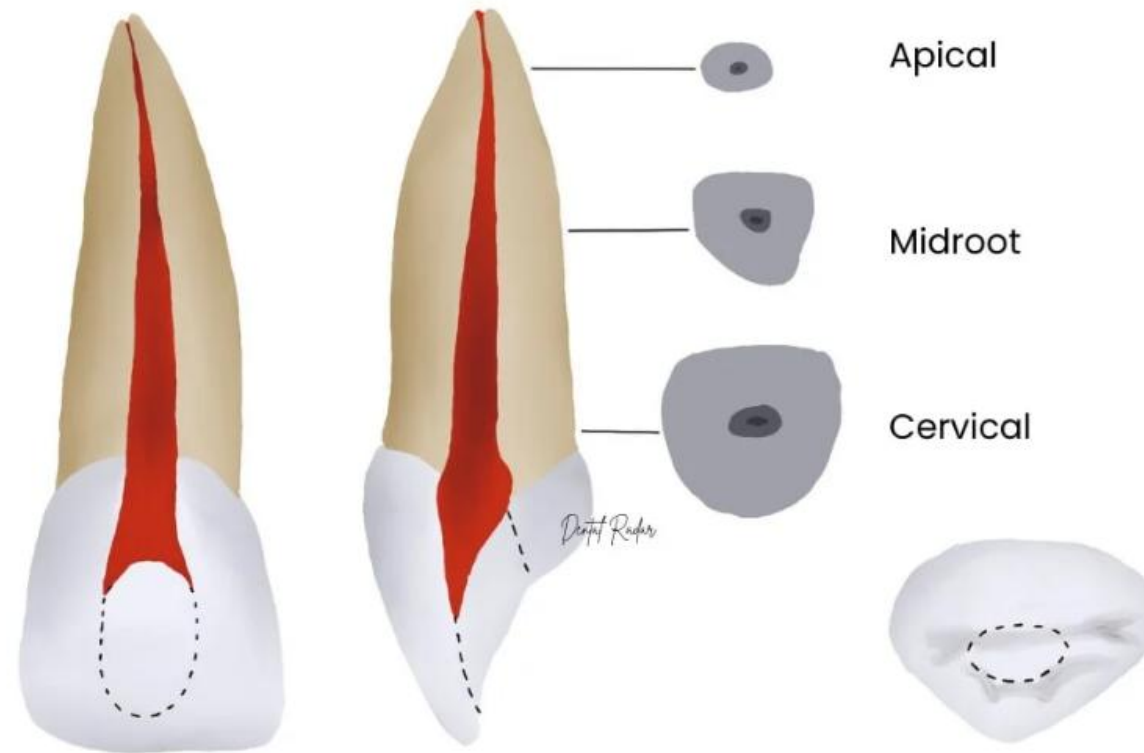


Fig. 2.60 Schematic representation of the pulp morphology of the maxillary first permanent incisor.



*Dotted lines shows outline of access cavity

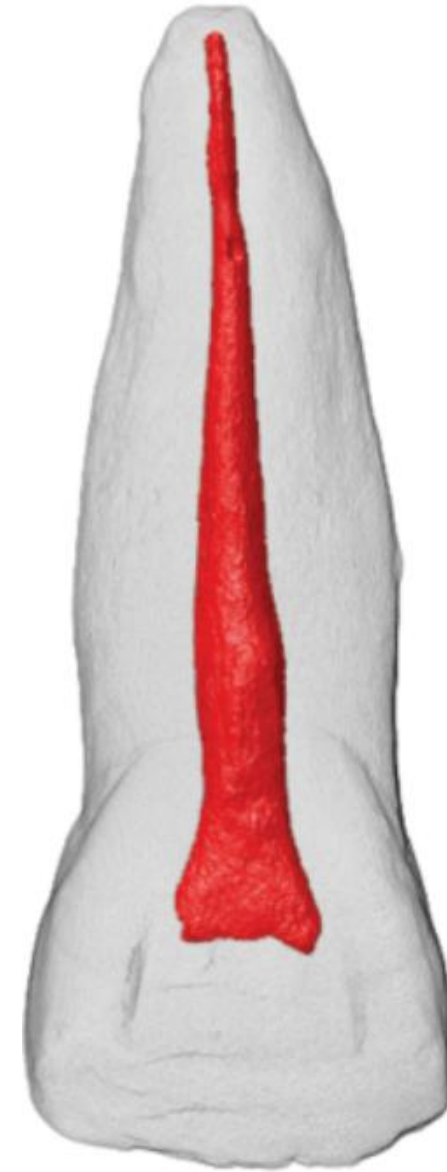
Labiolingual Section

- Pulp cavity follows crown/root outline.
- Pulp chamber is narrow incisally and widest cervically.
- Root canal tapers to the apical constriction.
- Apical foramen is usually near the tip, sometimes slightly labial or lingual.
Clinical: Root filling should end ~1 mm short of the radiographic apex (or use an apex locator).



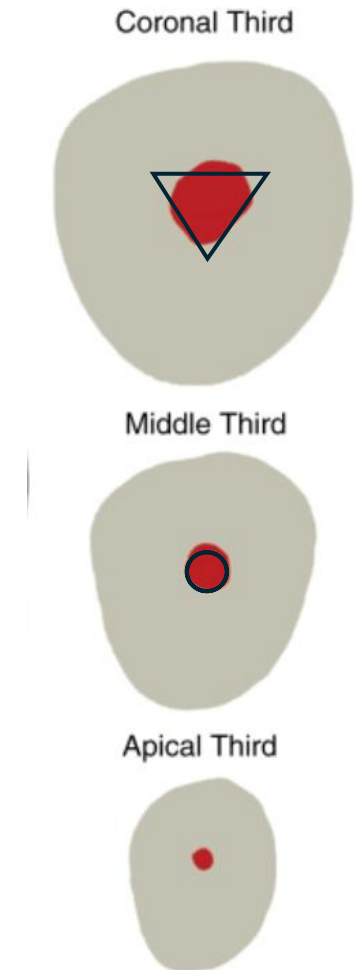
Mesiodistal Section

- Pulp chamber is **wider mesiodistally** than labiolingually.
- Pulp horns present if mamelons are prominent.
- Even taper along length; apical foramen often off-center.

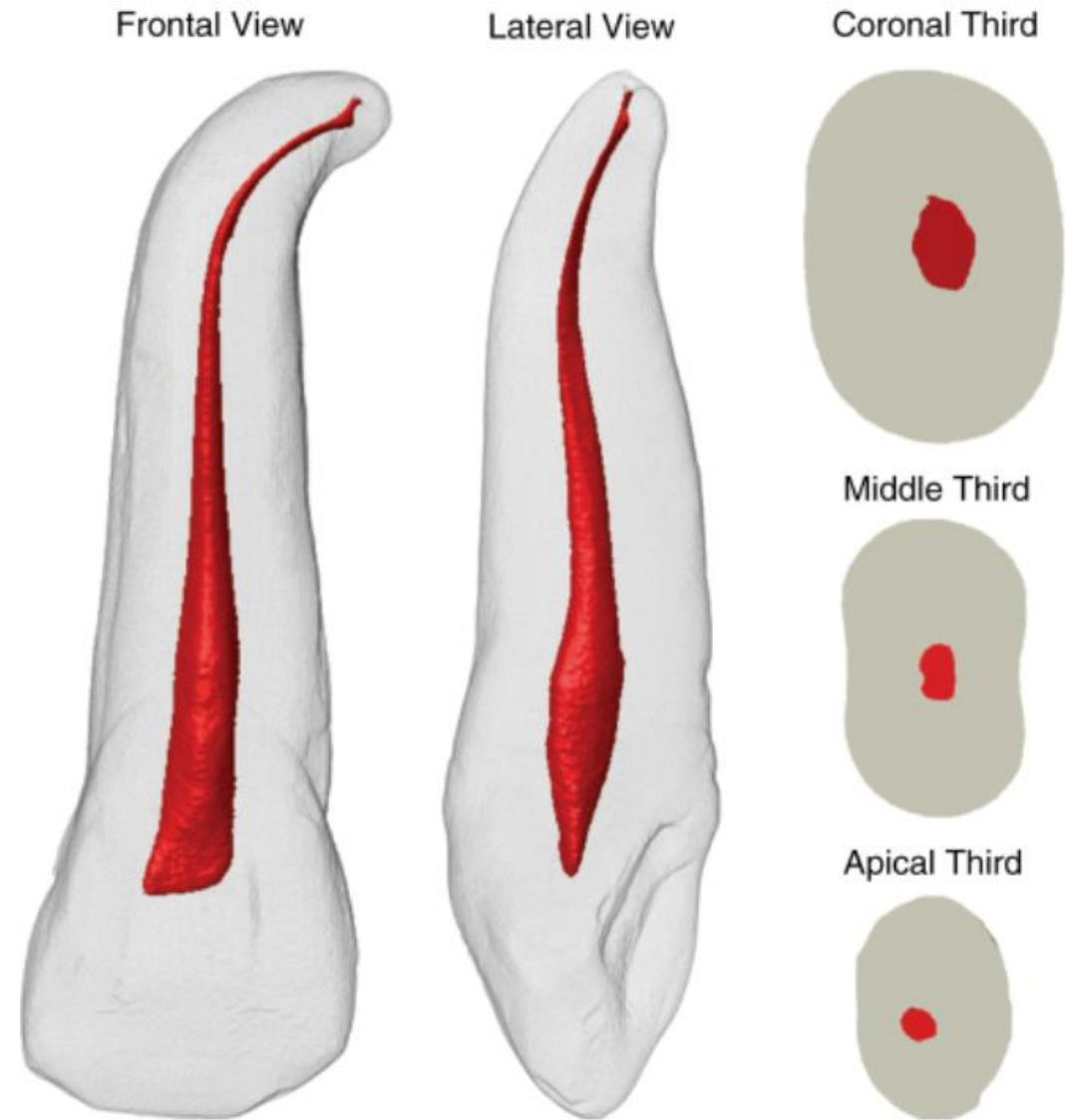


Cross Sections (Cervical & Midroot)

- Pulp cavity widest at cervical level; centered within root.
- **Cervical:**
 - ✓ Young – triangular (base labial).
 - ✓ Older – round/crescent due to secondary dentin.
- **Midroot:** Rounder, smaller but similar anatomy.



Maxillary Lateral Incisor



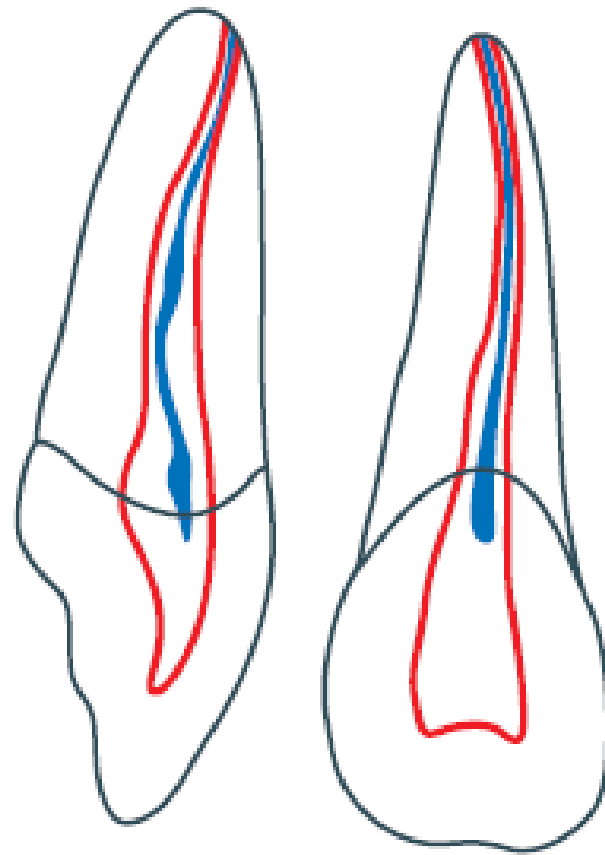


Fig. 2.62 Schematic representation of the pulp morphology of the maxillary second permanent incisor.

Labiolingual Section

- Similar to central incisor; but smaller.
- Pulp horns are usually prominent.
- **Pulp chamber: narrow incisally; may be very wide at cervical level** (some teeth lack this widening and taper evenly).
- **Apical foramen: often at tip, or may exit labially or lingually.**

Lateral View



Lateral incisor

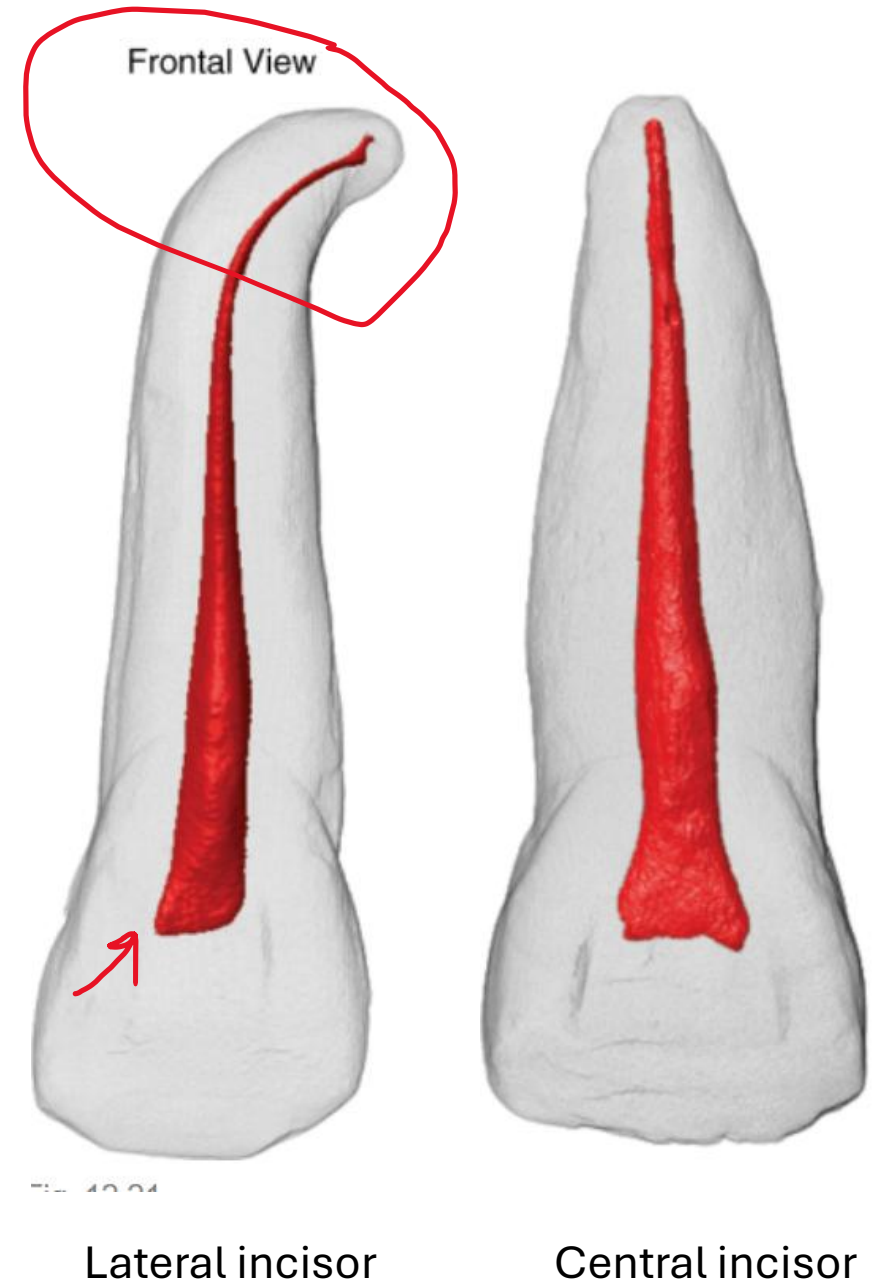
Lateral View



Central incisor

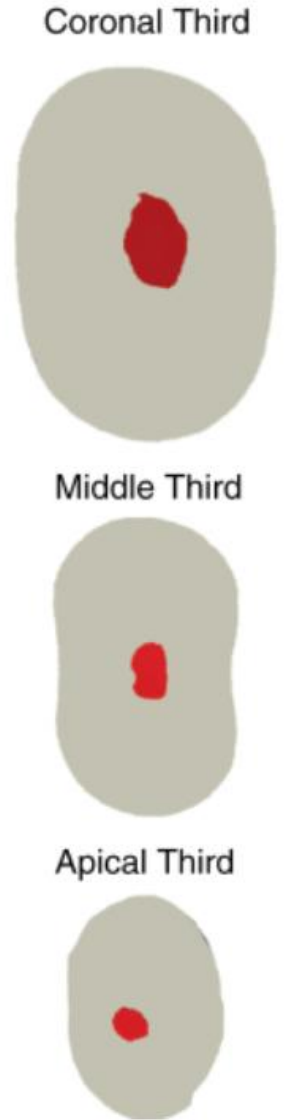
Mesiodistal Section

- Pulp horns appear blunted.
- Gradual taper toward apex;
- Apical region often shows a **significant distal or palatal curve**.
- **Clinical Note:** The distal and palatal curvature should be anticipated during endodontic instrumentation.

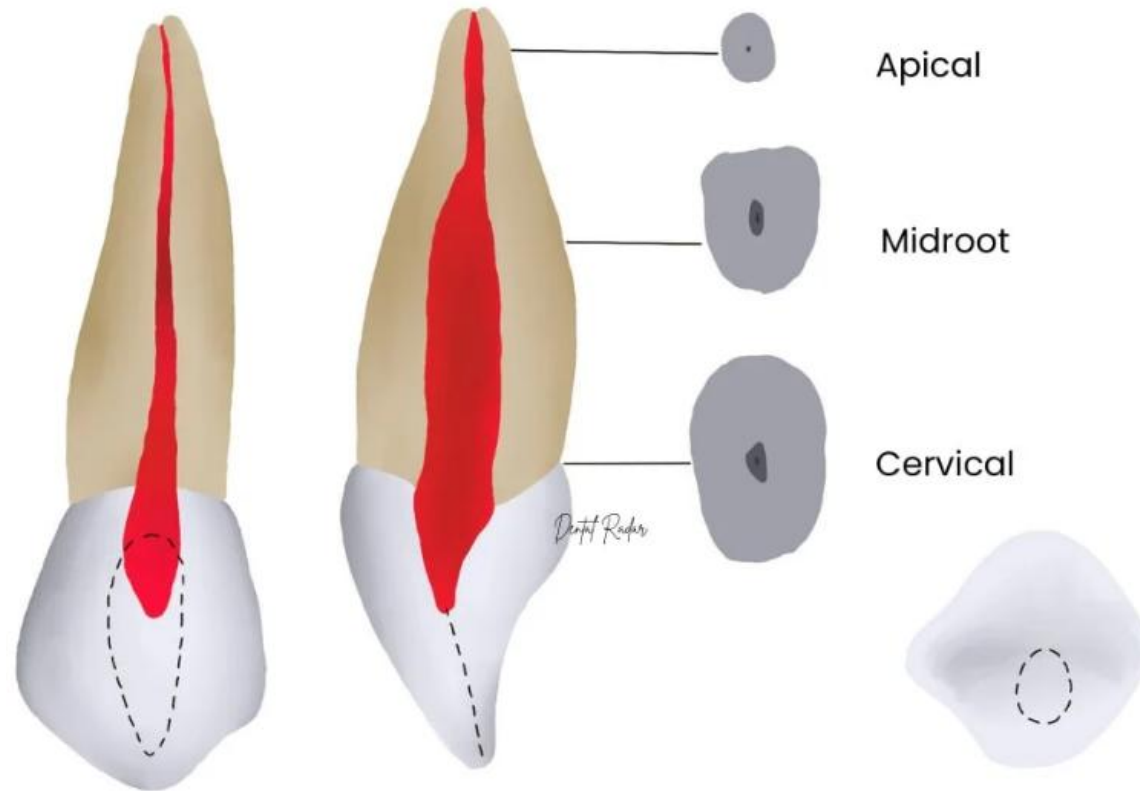


Cross sections (cervical & midroot):

- Pulp chamber is centrally located.
- Root shape varies (triangular, oval/ovoid, or round).
- Secondary dentin may significantly narrow the canal.



Maxillary Canine



*Dotted lines shows outline of access cavity

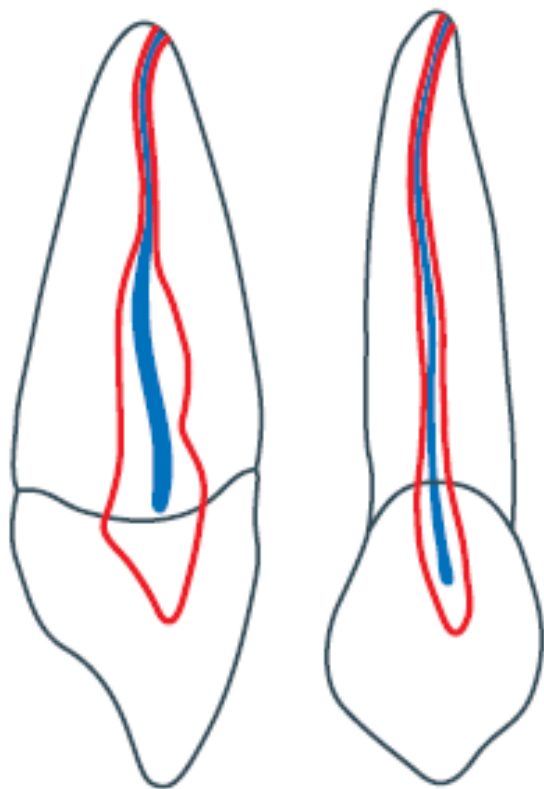
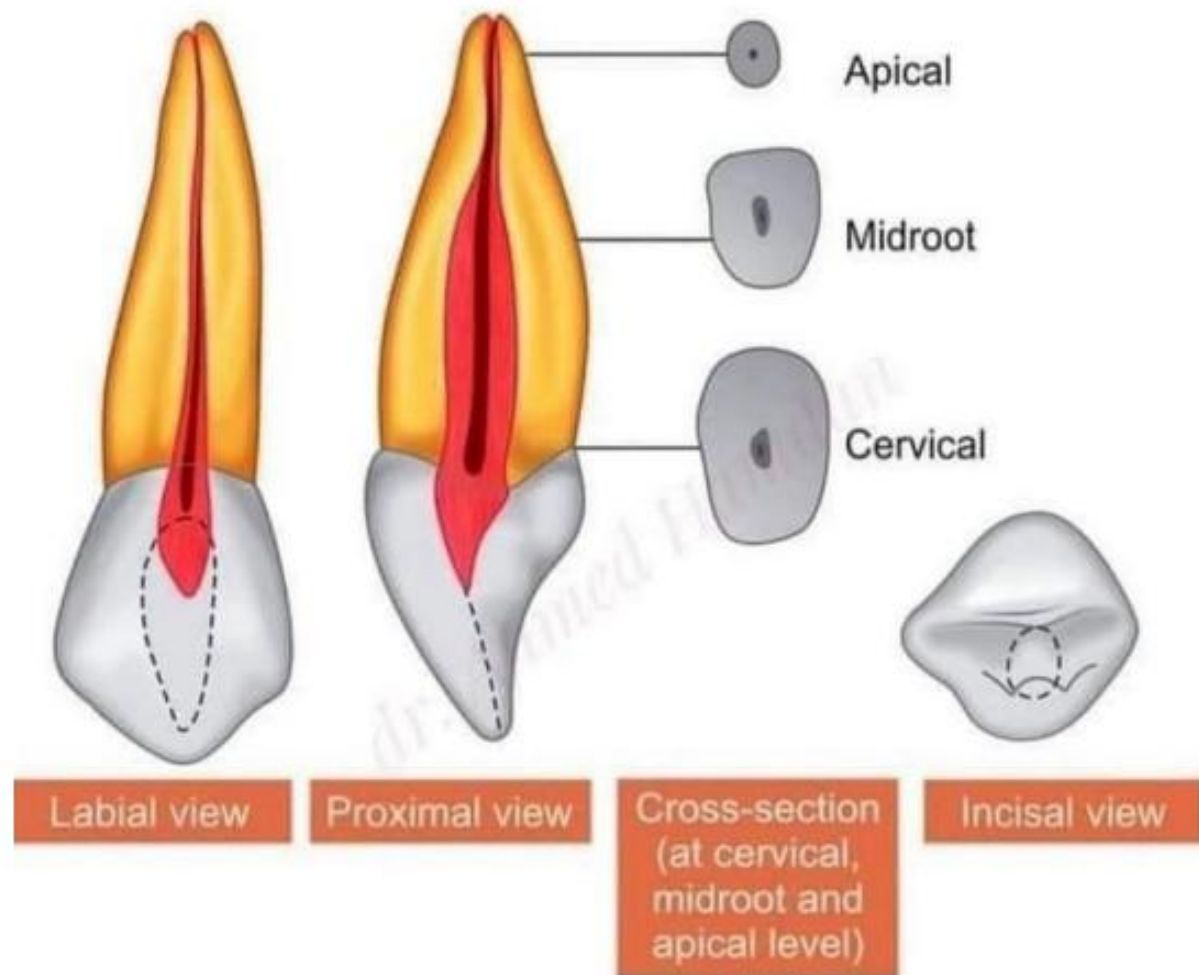


Fig. 2.66 Schematic representation of the pulp morphology of the maxillary permanent canine.



Dotted line shows the outline for access cavity preparation

Fig. 3.37 Maxillary canine

Labiolingual Section

- Wider **labiopalatally** than mesiodistally (both chamber and canal)
- **Largest pulp chamber**, Prominent pulp horn beneath cusp.
- **Root canal: Always single, with two taper patterns:**
 - ✓ gentle taper → wide incisally, then abrupt apical constriction
 - ✓ Even taper from chamber to apex
- Apical region may be sharply curved.
- Apical foramen at or slightly labial to apex.



Mesiodistal Section

- Pulp cavity much narrower than labiolingual view.
- Similar taper to incisors but with a **longer root**.
- Apical region may curve mesially or distally.
- Apical foramen: at tip or slightly mesial/distal.

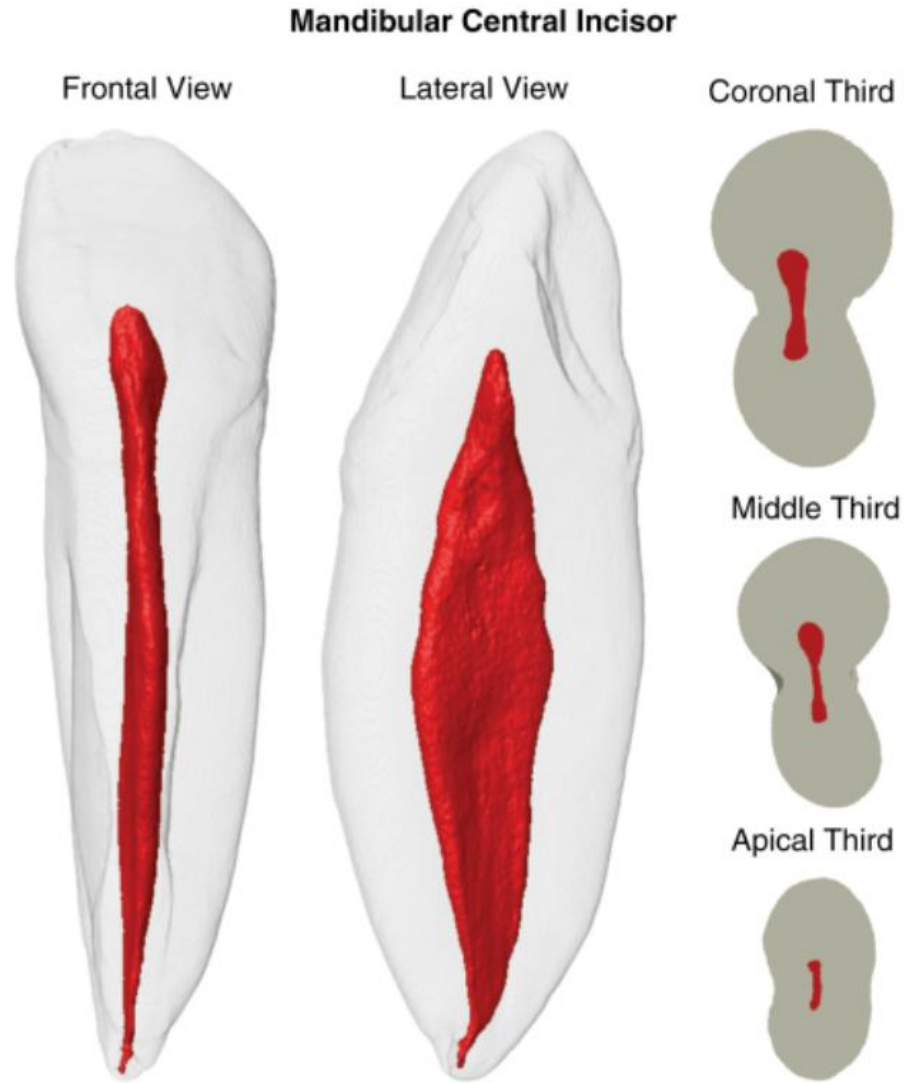


Cervical Cross Section

- Root and pulp chamber shape:
 - ✓ At mid-root and cervical part = oval, triangular, or elliptical
 - ✓ At apical third=round



Mandibular Central Incisor



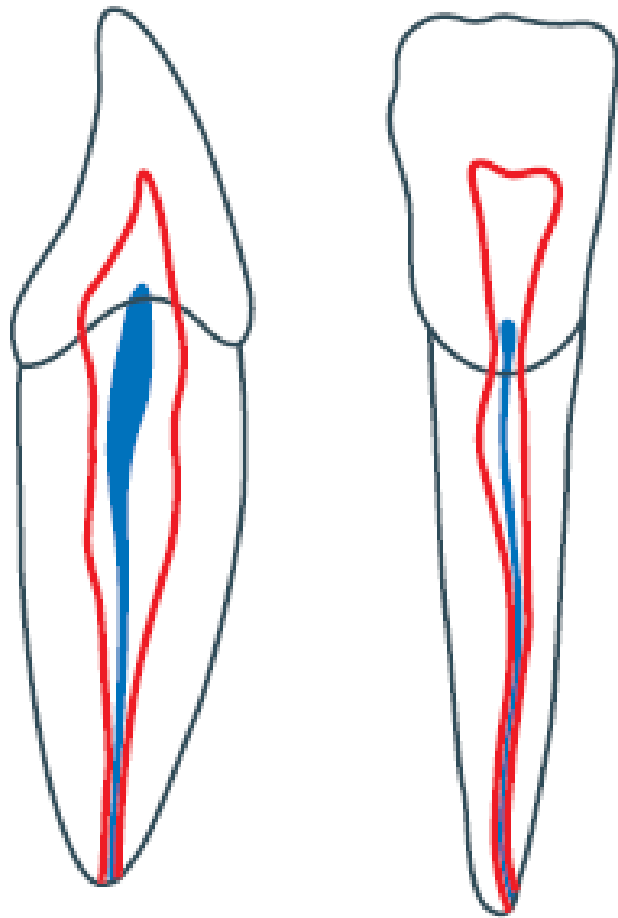


Fig. 2.63 Schematic representation of the pulp morphology of the mandibular first permanent incisor.



Labiolingual Section

- Pulp chamber similar to maxillary central incisor but smaller, with relatively large labiolingual dimension.
- **Cervical margin:** Shows a **constriction**.
- Well-developed single, prominent pulp horn.
- Canal tapers gently or shows abrupt narrowing in apical 3–4 mm (apical constriction).
- Apical foramen at apex or slightly buccal.

Lateral View



Labiolingual Section

- ✓ Usually one canal.
- ✓ Two canals: Present in up to 30% of mandibular central incisors.
 - Most fuse near the apex and exit through a single foramen.

Clinical Implication: Always suspect a second canal
careful exploration is essential.



Mesiodistal Section

- Pulp cavity appears very narrow (small mesiodistal width), compensated by wide labiolingual dimension (aids file access in cavity preparation).
- Secondary/tertiary dentin may complicate treatment as attrition is common.
- Gentle taper of root canal to apex.
- Apical foramen at apex or slightly mesial/distal.

Frontal View



Cervical Cross Section

- Mesiodistal dimension = small;
- labiolingual dimension = very large.
- Root canal shape: ovoid/round/elliptical then round in apical third
- Two canals possible (rare); dentinal island may mimic two canals.

Coronal Third



Middle Third



Apical Third



Mandibular Lateral Incisor

Mandibular Lateral Incisor

Frontal View



Lateral View



Coronal Third



Middle Third



Apical Third



- **Two canals:** Somewhat more common – present in approximately 43%.
- **Foramina:** When two canals are present, most exit by separate foramina (unlike central incisor where they often fuse).

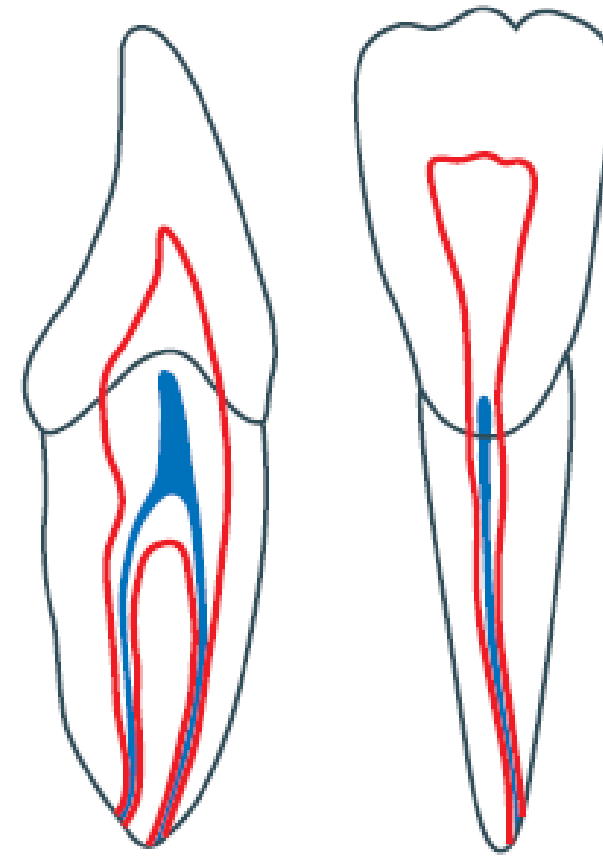


Fig. 2.65 Schematic representation of the pulp morphology of the mandibular second permanent incisor.

Labiolingual Section

- Slightly larger than central incisor in all dimensions; pulp chamber also larger.
- Pulp horn: prominent.
- Root canal taper: gentle or abrupt with narrowing in last 3–4 mm.
- Apical foramen: at tip or buccal/lingual aspect.



Mesiodistal Section

- Slender cavity; resembles central incisor but slightly wider with larger pulpal dimension.
- Prominent pulp horns.
- Gentle taper to apex.
- Apical foramen: at tip or mesial/distal aspect.



Cervical Cross Section

- Pulp canal centered in root.
- Root larger than mandibular central incisor.
- Root shape: oval to elliptical (considerable variation).
- Some roots demonstrate root grooves.

Coronal Third



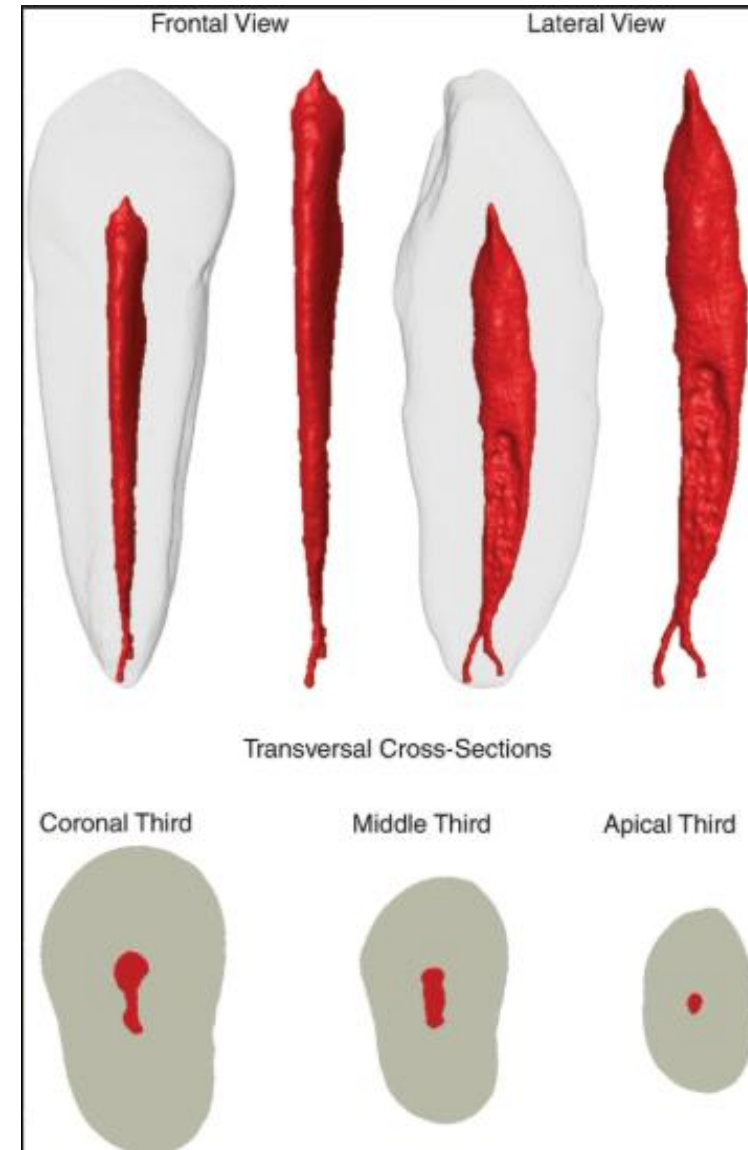
Middle Third



Apical Third



Mandibular Canine



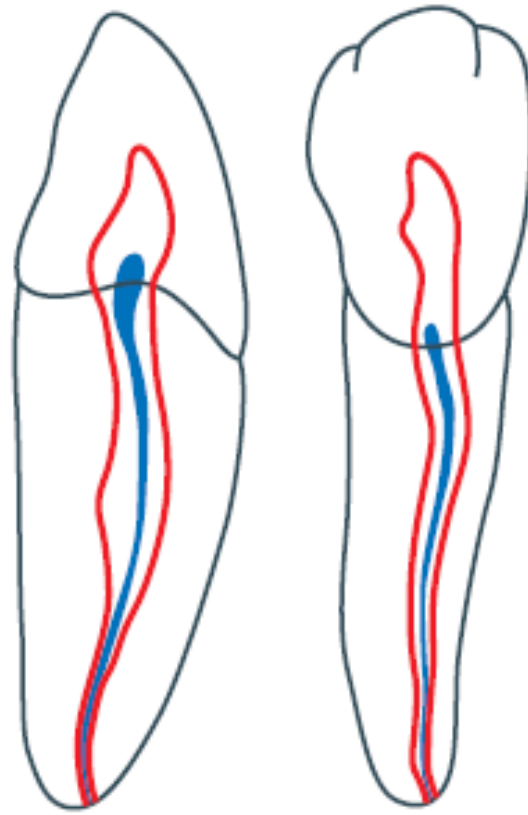


Fig. 2.68 Schematic representation of the pulp morphology of the mandibular permanent canine.

Labiolingual Section

- Similar to maxillary canine, but usually shorter.
- Two canals (or roots) are not uncommon.
- Dentinal island may be present with wide labiolingual, narrow mesiodistal dimension .
- Two canals may not be visible radiographically but can confirm clinically.
- Prominent pulp horn (unless attrition).
- Pulp chamber typically wide.
- Taper patterns: abrupt from chamber, abrupt apically then gentle, or gentle throughout.
- Apical foramen may be present at tip, buccal, or lingual.



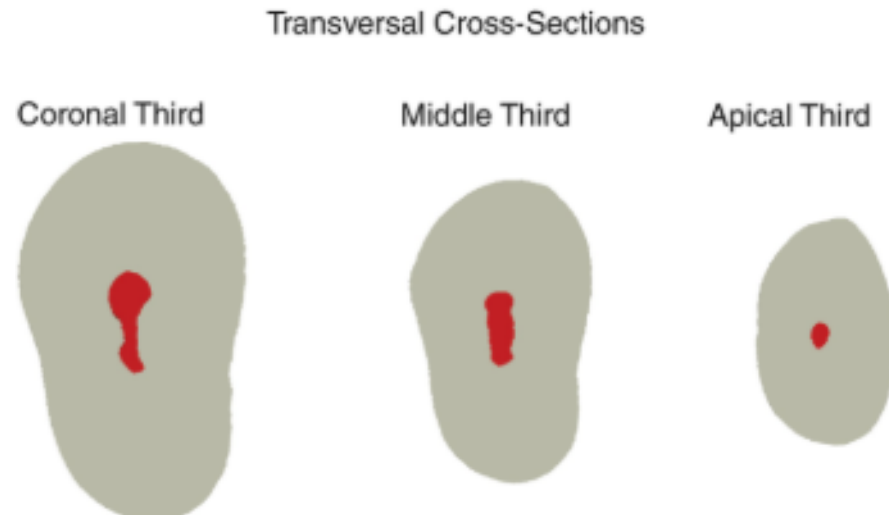
Mesiodistal Section

- Similar to maxillary canine; narrow mesiodistal width.
- Apical region may curve mesially.
- Pulp horn appears blunted.
- Gentle taper to the apex.
- Apical foramen at tip or slightly mesial/distal.

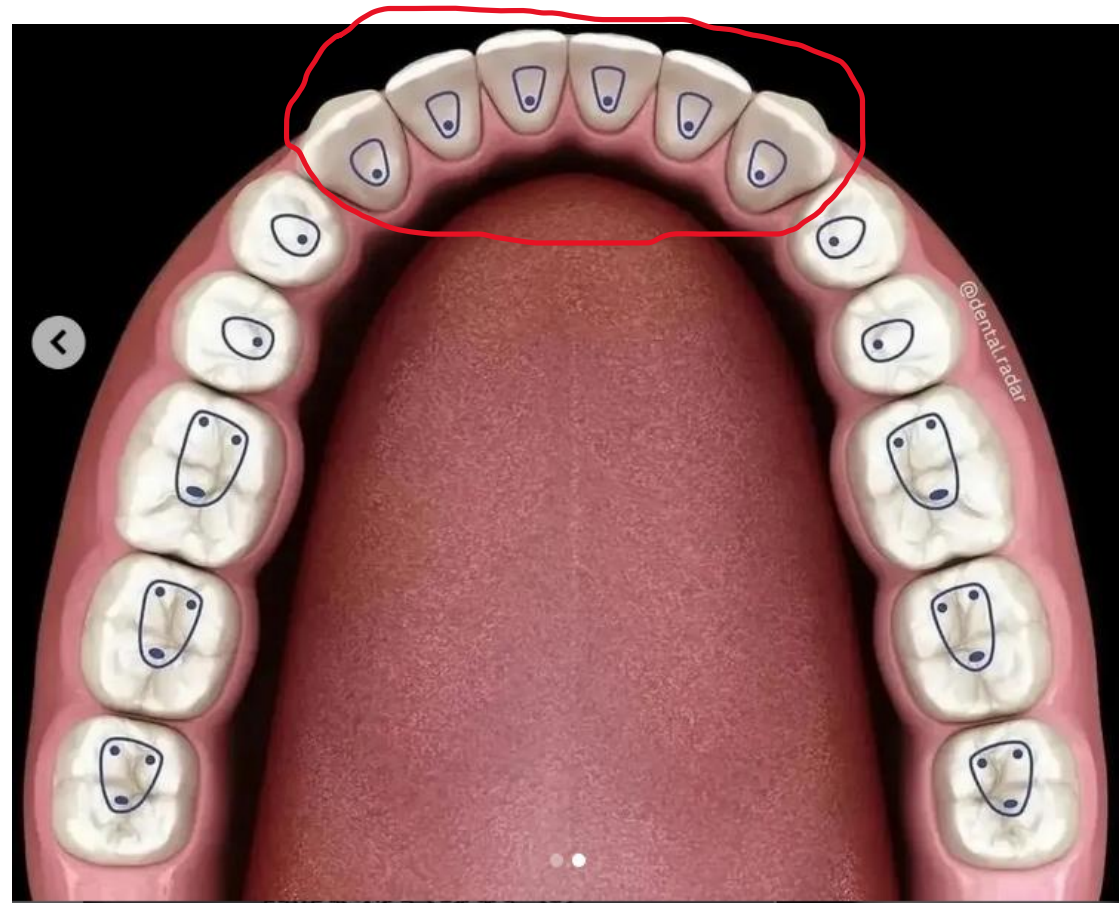
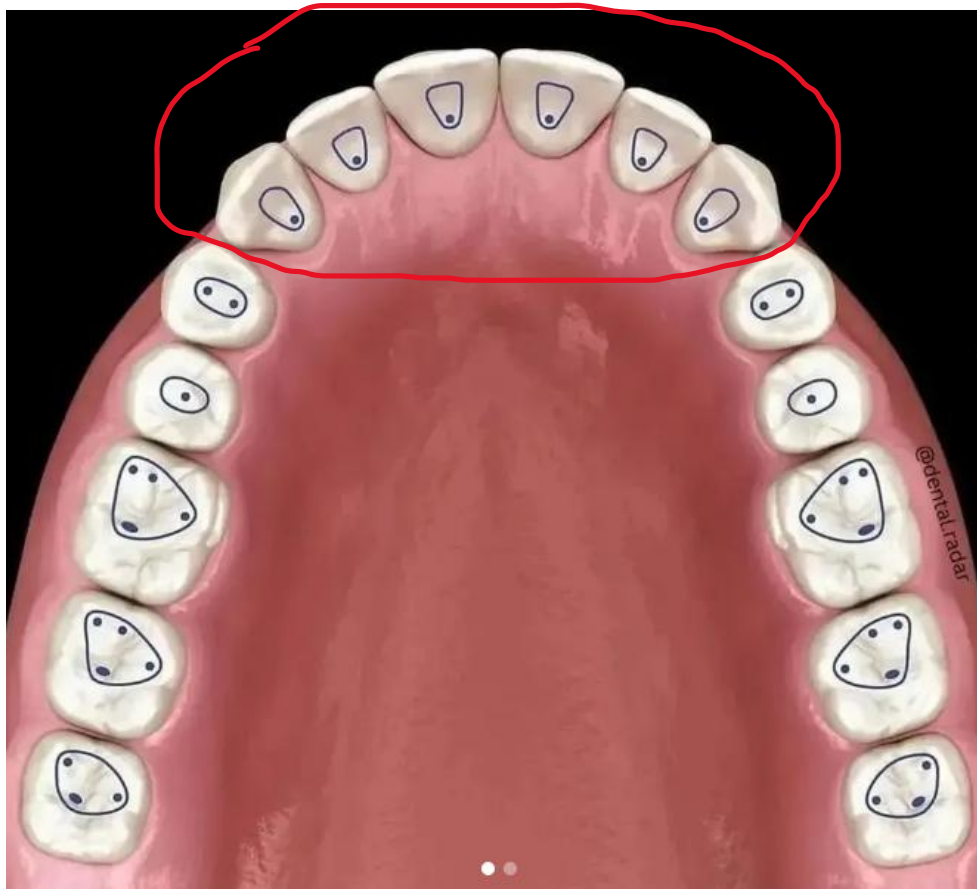


Cervical cross section:

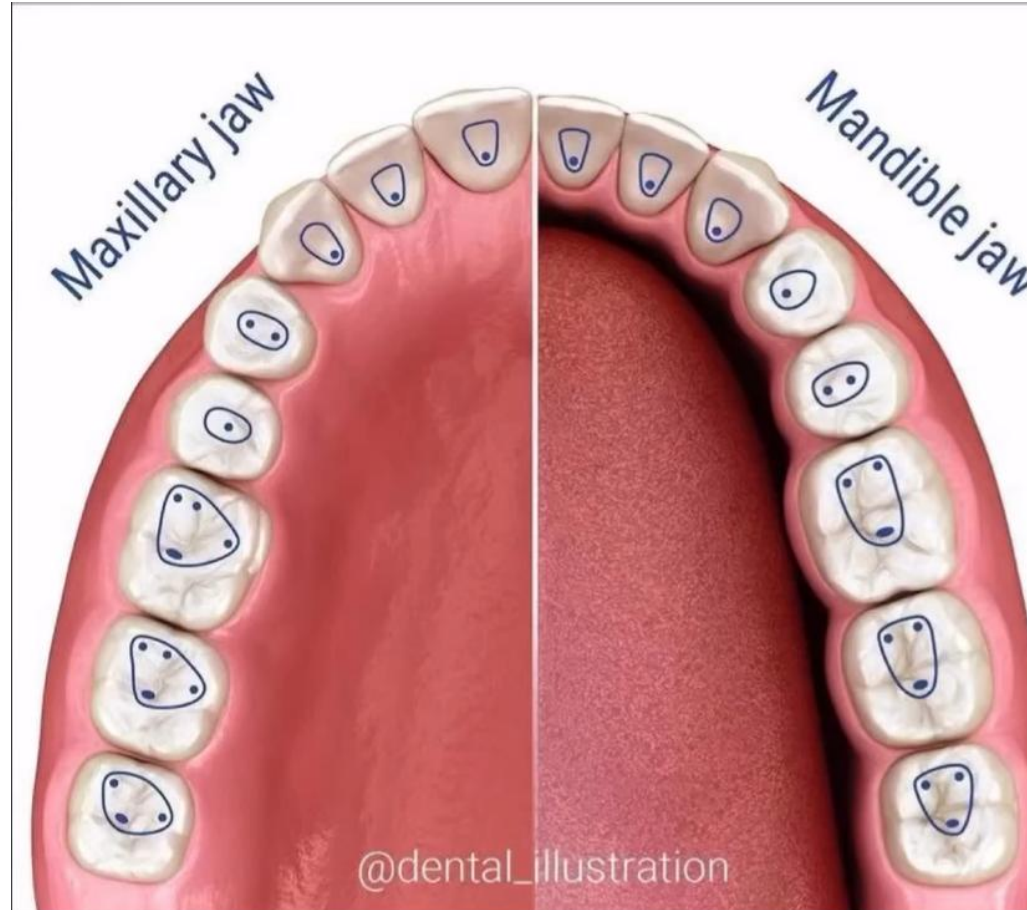
- Considerable variation in size and shape.
- Root outline may be oval, rectangular, or triangular.
- Pulp cavity closely follows the root form.



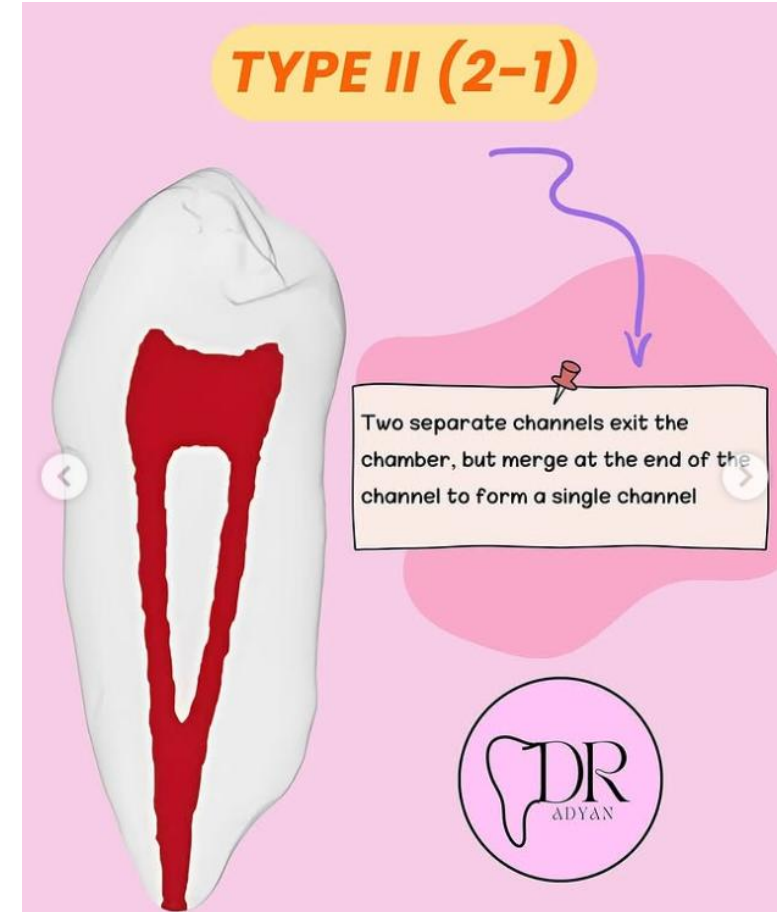
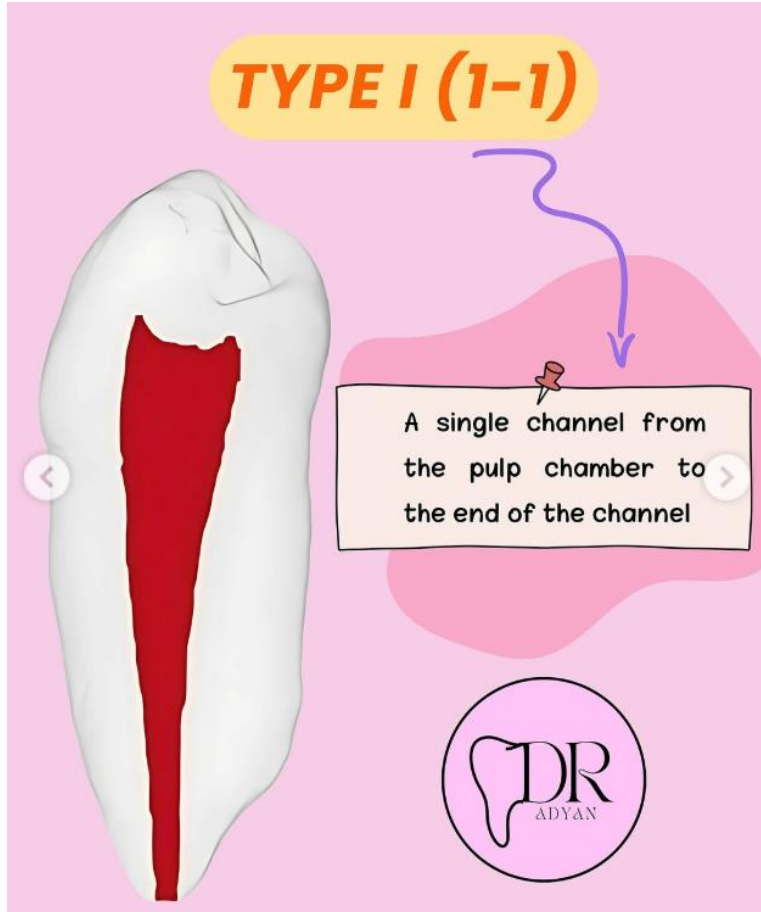
The shape and position of the access cavity of the teeth in the upper and lower jaw.



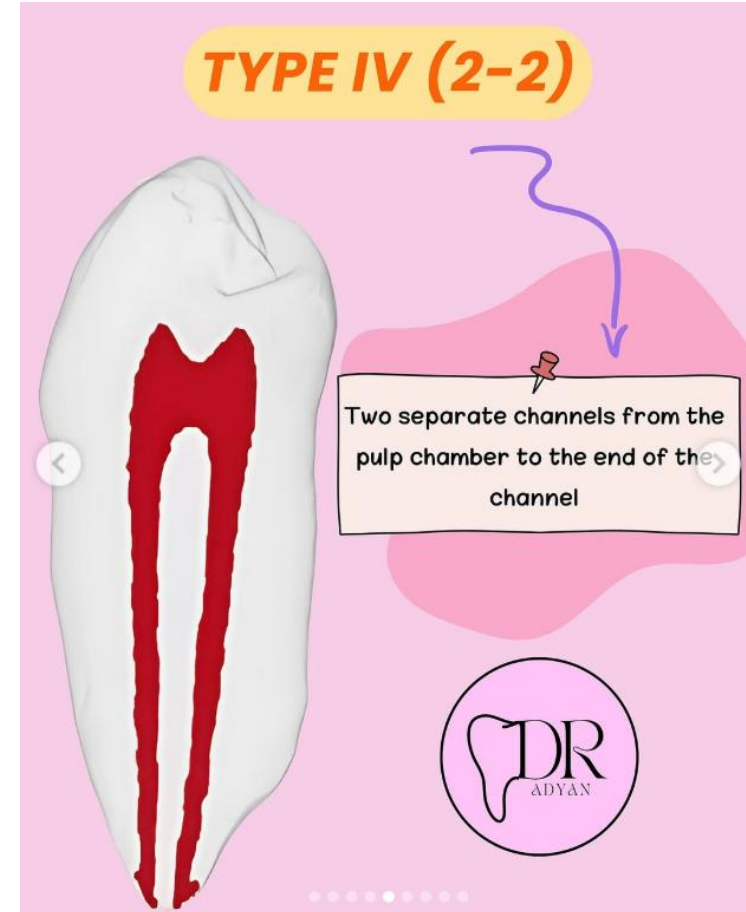
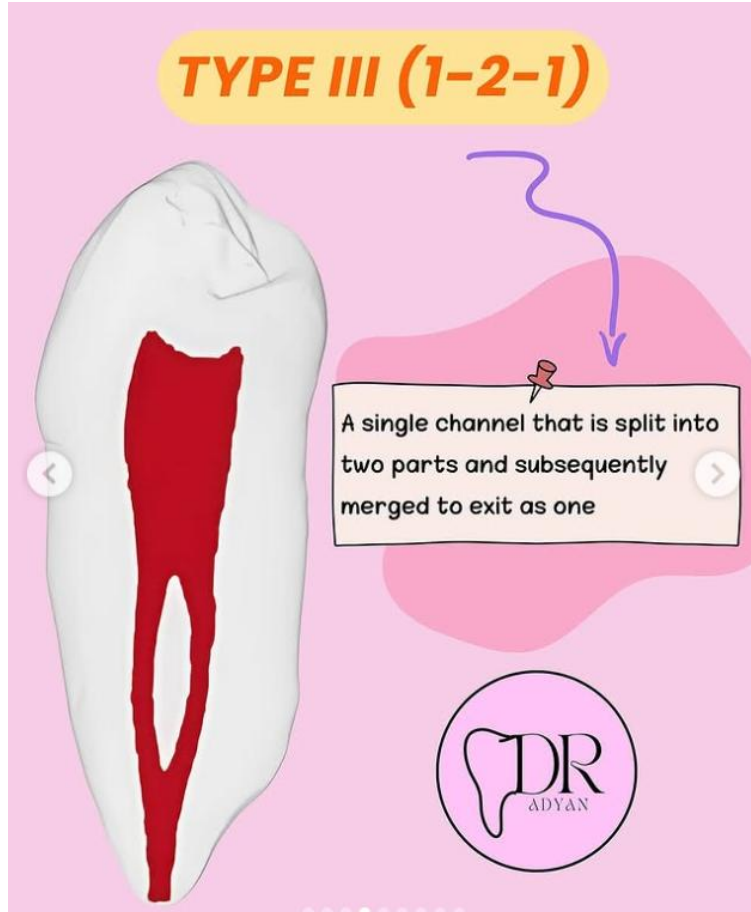
Root canal orifice location plan of the maxillary and mandibular jaw



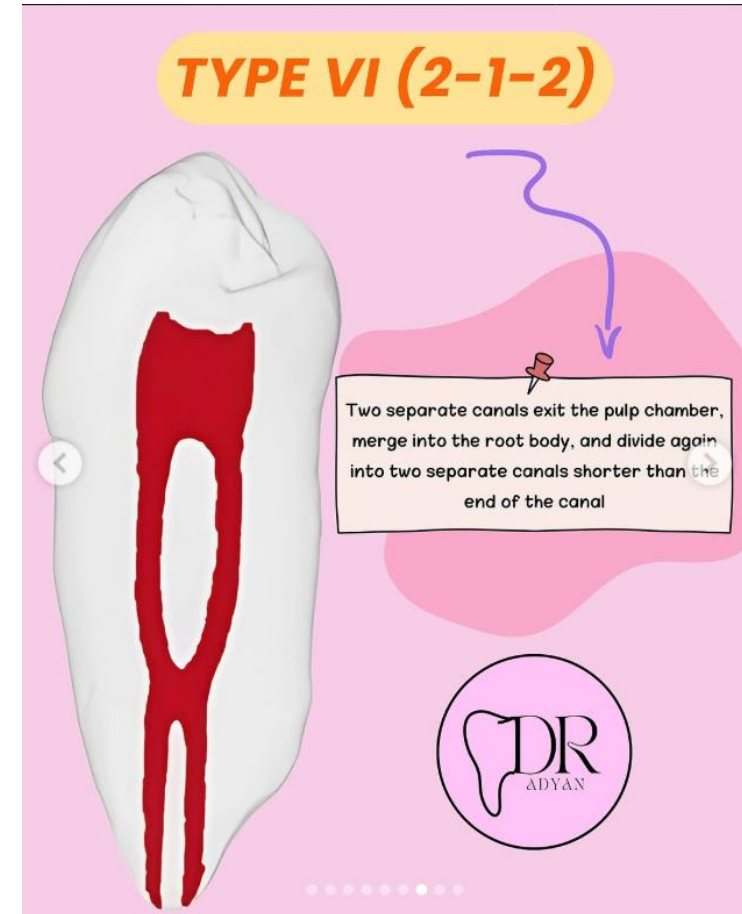
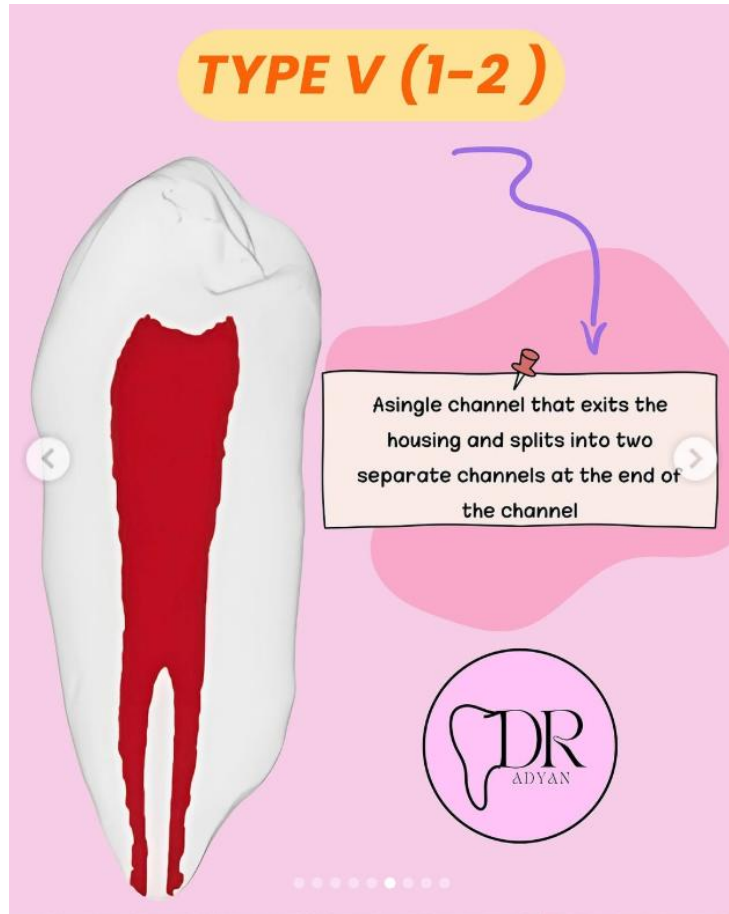
Vertucci's classification for root canal morphology



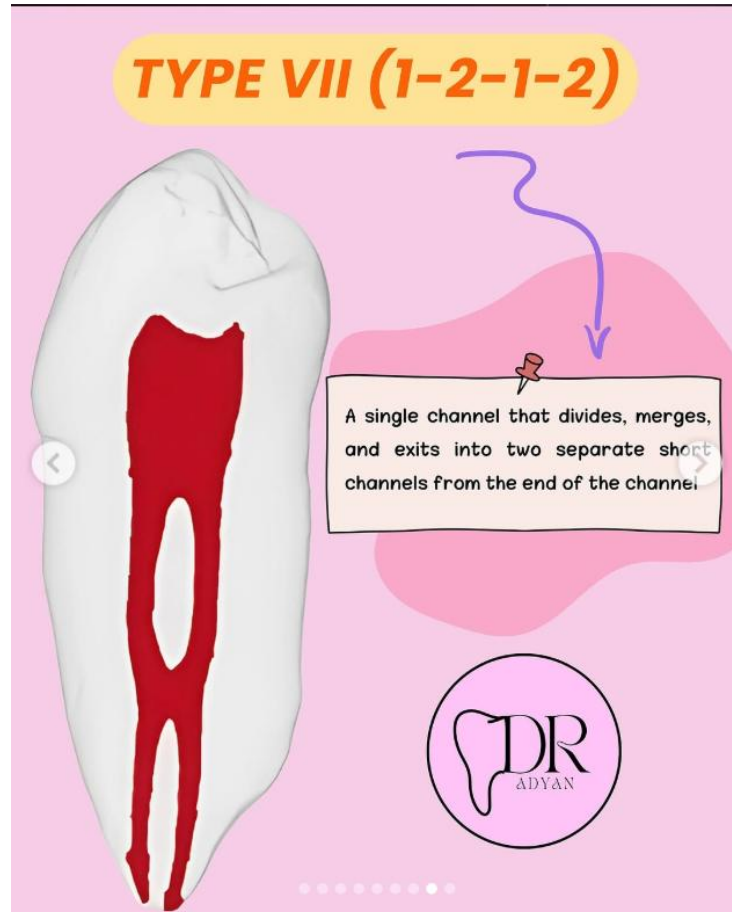
Vertucci's classification for root canal morphology



Vertucci's classification for root canal morphology

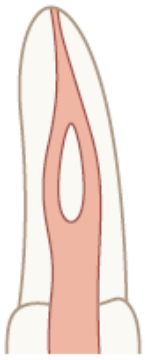


Vertucci's classification for root canal morphology





Type I



Type II



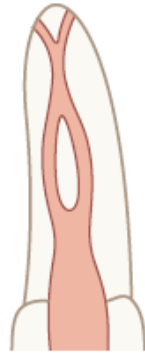
Type III



Type IV



Type V



Type VI



Type VII



Type VIII

Fig. 2.58 Types of root canal configuration.

Shape of the pulp chamber in deciduous anterior teeth



Key Differences from Permanent Teeth

- **General principle:** Pulp chamber shape reflects crown shape, but in deciduous teeth:
 - Pulp chamber is relatively larger.
 - Pulp horns are longer and closer to the surface.

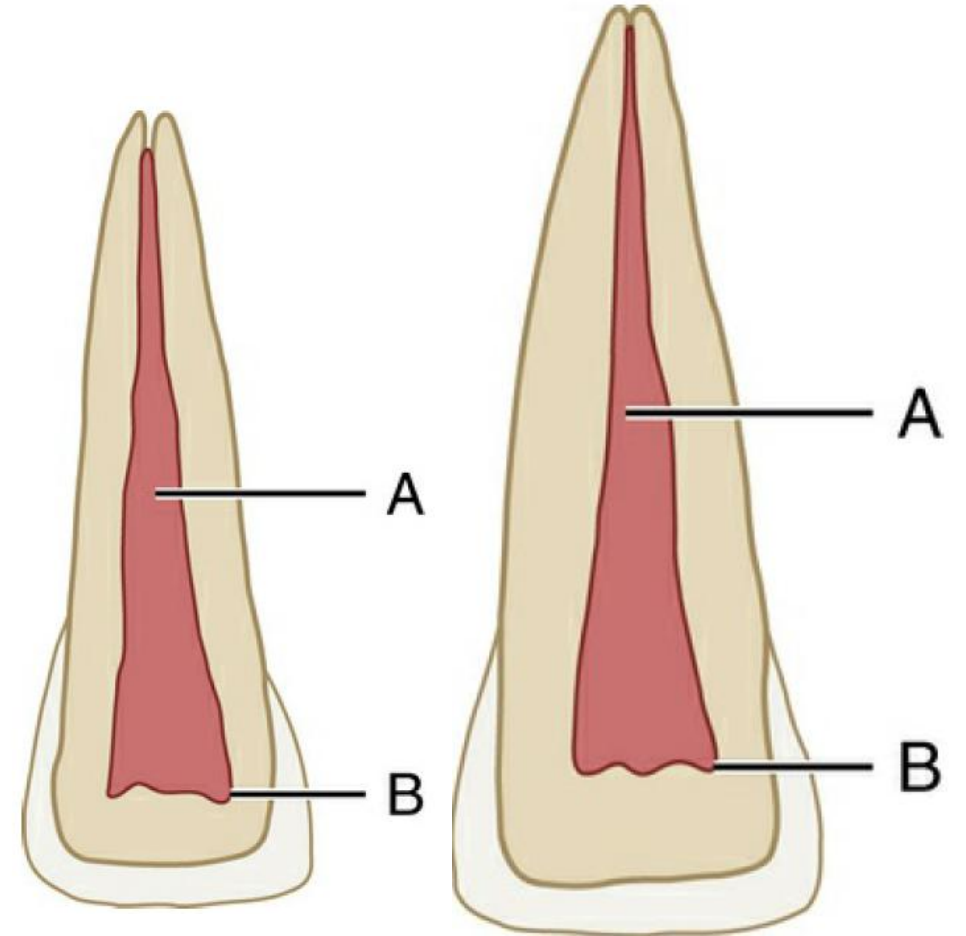


FIG. 3.9 Primary central incisor.

FIG. 3.8 Permanent central incisor.

- **Root canals:** All incisors and canines have **single canals**.
 - Cross-section: round or oval (compressed mesiodistally).
- **Two-canal variation:** Occurs in **10% of deciduous mandibular incisors**.
- **Clinical implication:** Higher risk of pulpal exposure during cavity preparation due to large pulp chamber and prominent pulp horns.

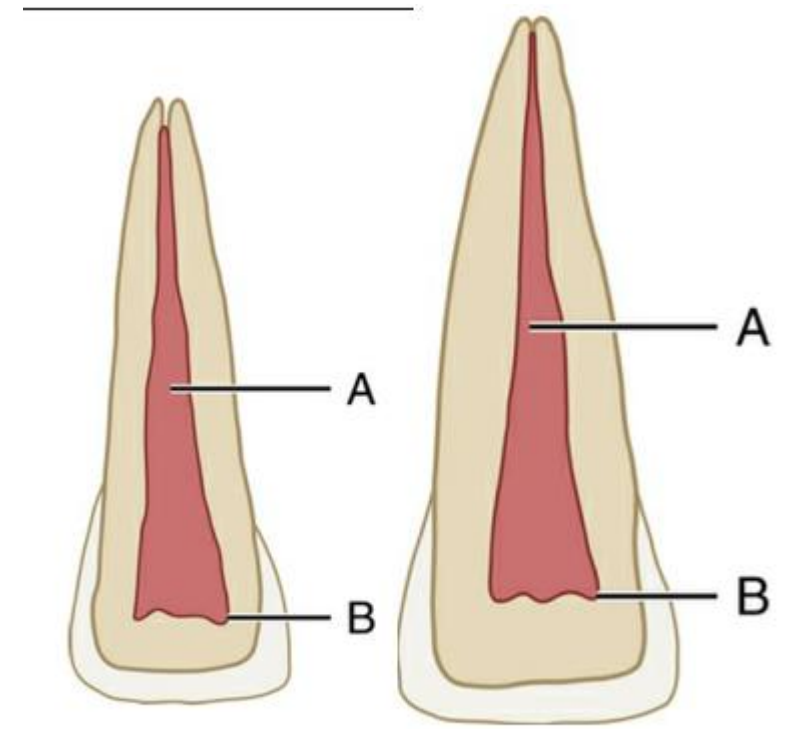


FIG. 3.9 Primary central incisor.

FIG. 3.8 Permanent central incisor.

A 22-year-old patient requires a class IV composite restoration on the maxillary central incisor. The preoperative radiograph shows a large pulp chamber with three distinct pulp horns extending into the incisal third. What is the most likely explanation for this finding?

- A. The tooth has been traumatized and has developed tertiary dentin
- B. The patient has a developmental anomaly resulting in extra pulp horns
- C. The tooth is young and the mamelons are still present
- D. The tooth has undergone significant attrition
- E. The pulp chamber has been reduced by secondary dentin deposition

A dental student is examining a radiograph of a maxillary lateral incisor and notices that the root canal curves distally in the apical third. Which of the following statements is true regarding this finding?

- A. It is a rare anomaly requiring advanced imaging
- B. The curve is always accompanied by a palatal curve
- C. It is a common anatomic variation in lateral incisors
- D. It indicates the presence of two canals
- E. It is caused by previous trauma

A 40-year-old patient has a mandibular lateral incisor with a complex root canal system. The clinician suspects two canals based on the tooth's morphology. Approximately what percentage of mandibular lateral incisors have two root canals?

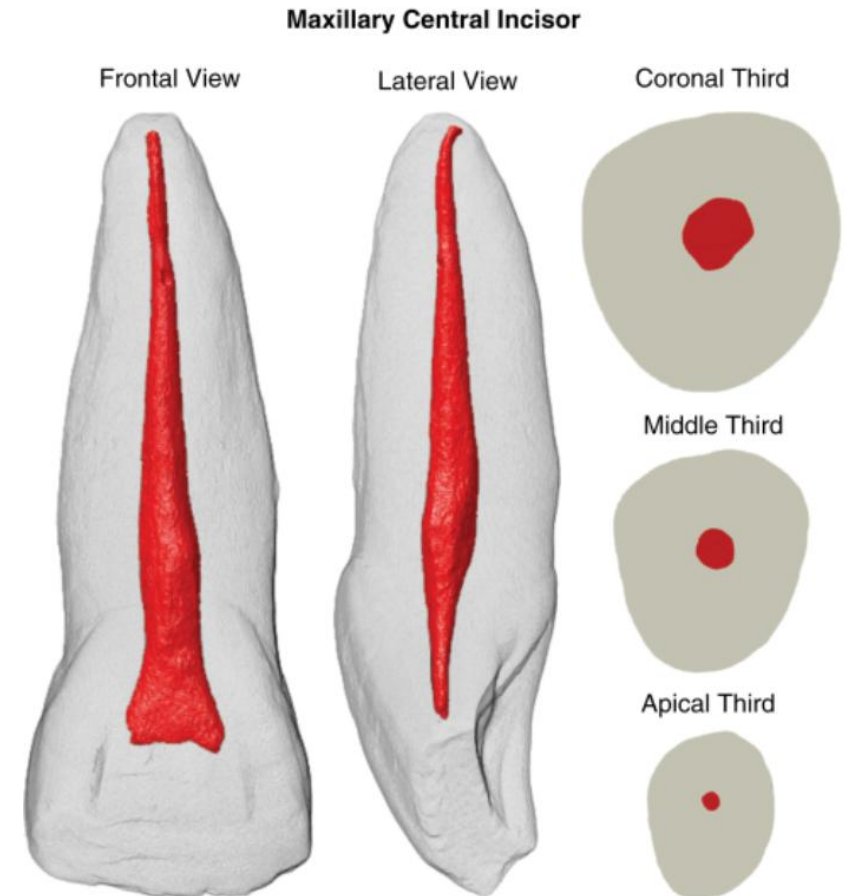
- A. 10%
- B. 30%
- C. 43%
- D. 6%
- E. 2%

Key
points



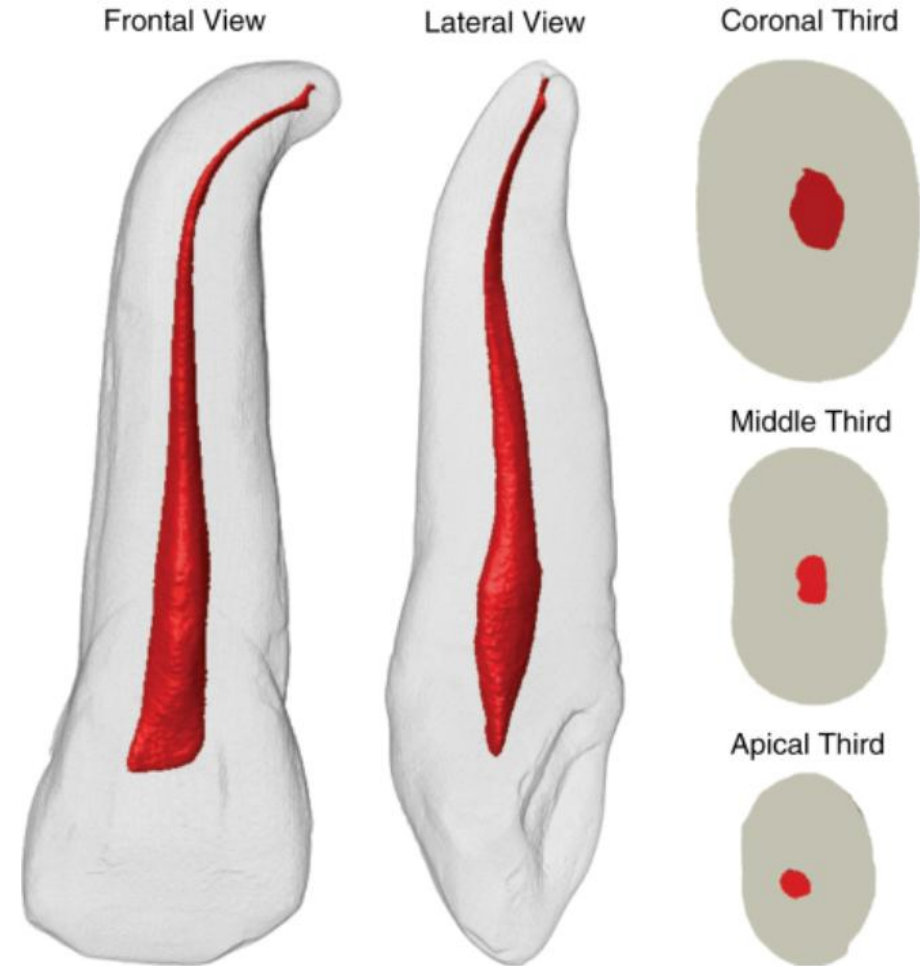
Maxillary Central Incisor – Pulp Morphology

- **Title:** Three Pulp Horns in Young tooth
- **Labial aspect:** Pulp chamber follows crown outline; widest toward incisal edge.
- **Young teeth: Three pulp horns** corresponding to mamelons.
- **Proximal view:** Pulp tapers incisally, widens cervically. **Cervical bulge** separates pulp chamber from single, centrally placed root canal.
- **Root canal:** Tapers to apex; may curve distally or labially.
- **Cross-section:** Ovoid for most of length; becomes round near apex.
- **Age changes:** Secondary dentin reduces chamber size; chamber may disappear completely → clinical challenge in locating canal.



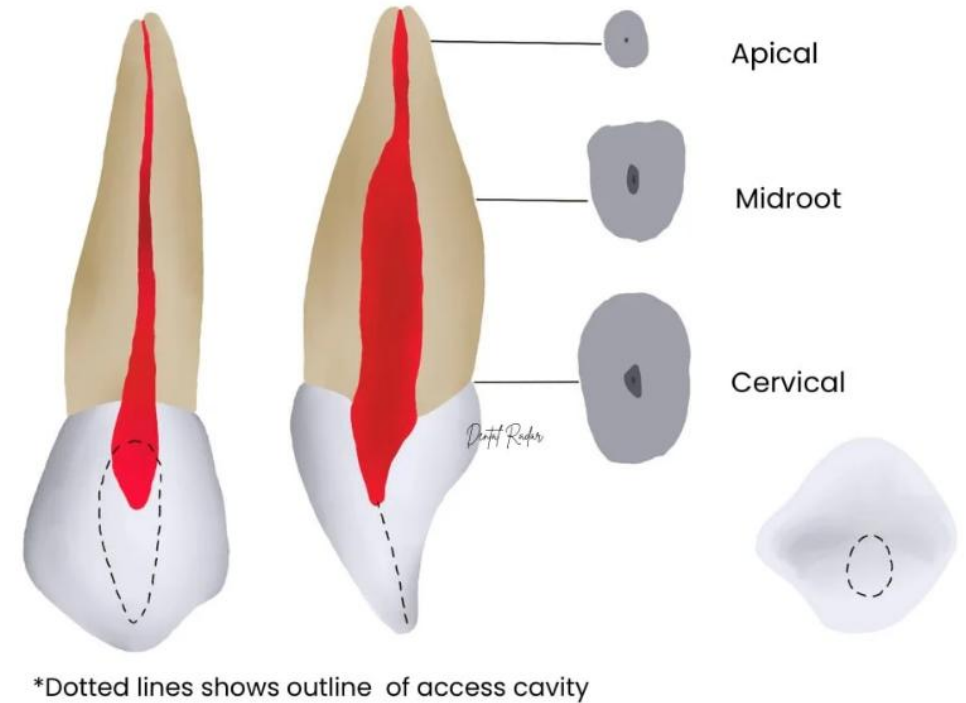
Maxillary Lateral Incisor – Pulp Morphology

- **Title:** Smaller, with Distopalatal Curve
- **Pulp chamber:** Similar to central incisor but **smaller**.
- **Root canal:** **Single**, slightly ovoid.
- **Characteristic curvature:** Commonly curves **distally** and **palatally**.
- **Clinical note:** Anticipate distal and palatal curvature during endodontic instrumentation.



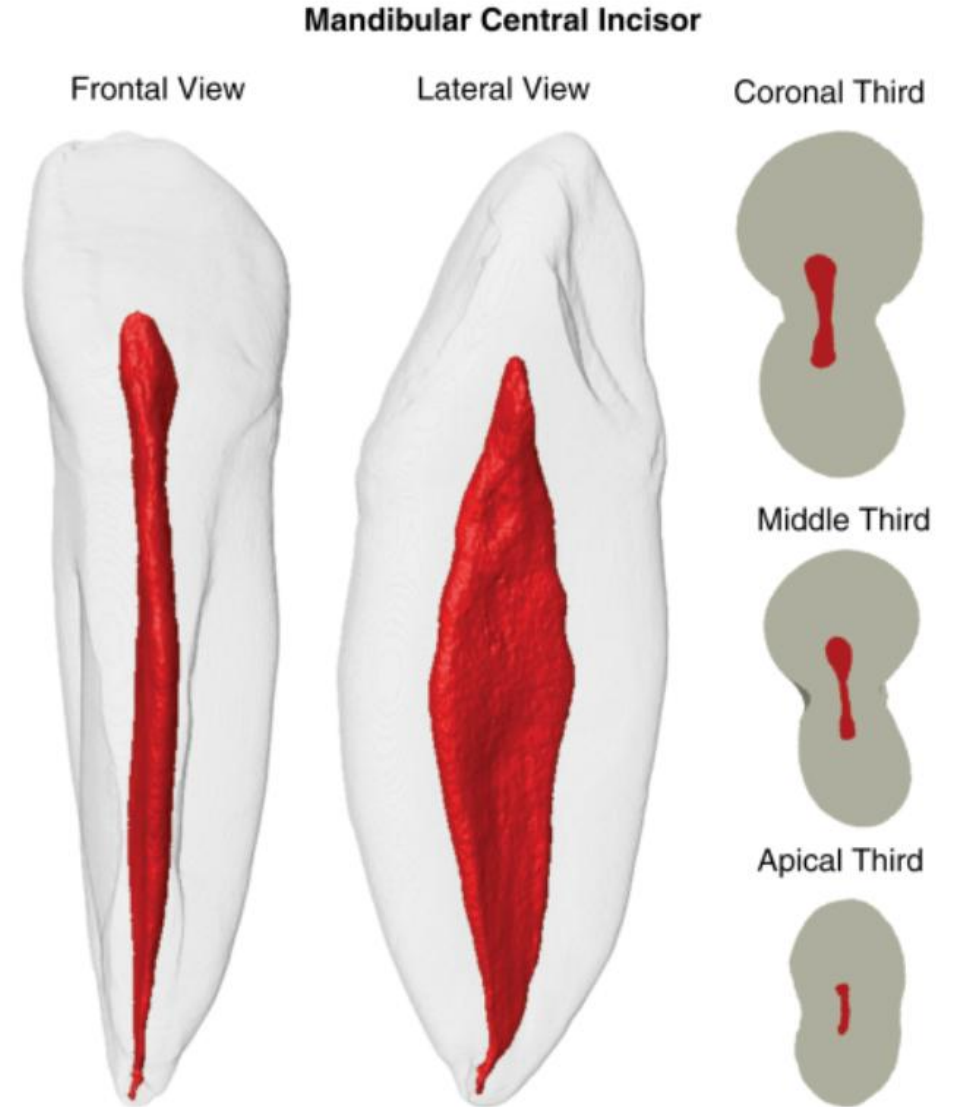
Maxillary Canine – Pulp Morphology

- **Title:** Narrow Chamber, Single Canal
- **Pulp chamber:** Narrow, with **single pulp horn** pointing cusally.
- **Shape:** Wider labiopalatally than mesiodistally.
- **Root canal:** Always single.
- **Constriction:** Does not narrow markedly until the **apical third**.
- **Cross-section:** Oval or triangular for most of length; **round** in apical third.



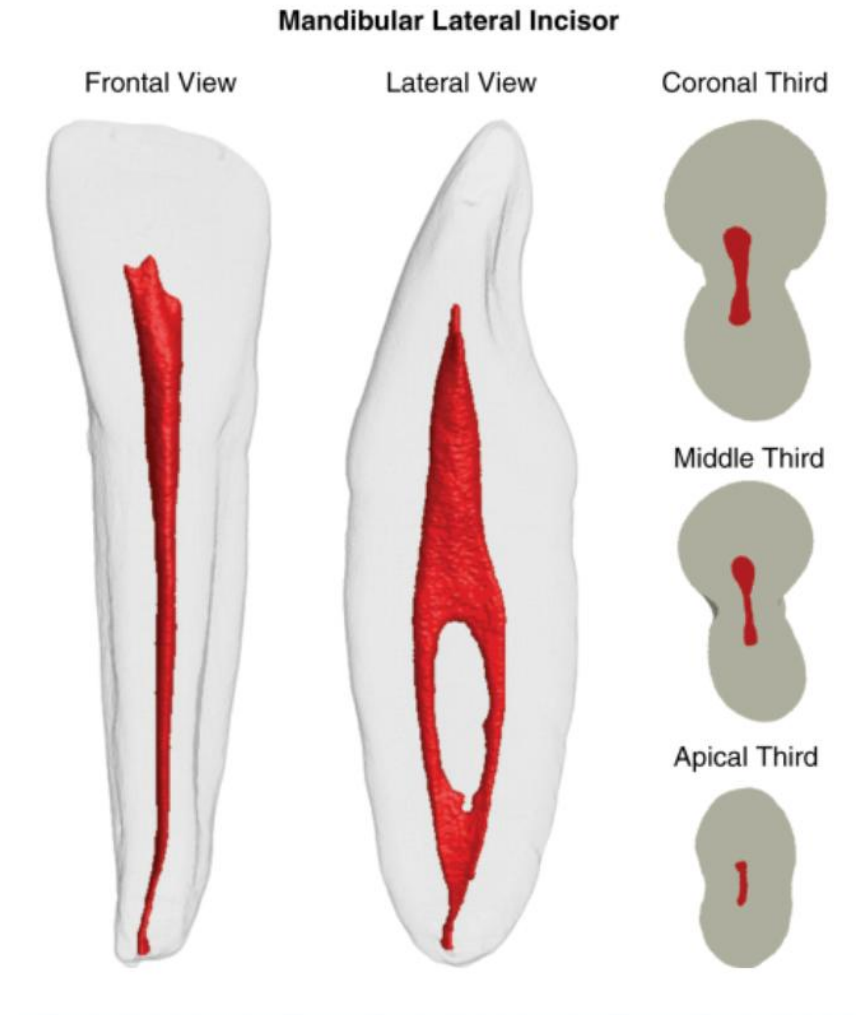
Mandibular Central Incisor – Pulp Morphology

- **Title:** Small, with Frequent Two-Canal Variation
- **Pulp chamber:** Similar to maxillary central but **smaller** (tooth is smaller).
- **Cross-section:** Oval, wider labiolingually than mesiodistally.
- **Cervical constriction:** Present.
- **Root canal:** Ovoid → round in apical third.
- **Two canals:** Present in up to **30%**.
 - Most **fuse near apex** and exit through a **single foramen**.
- **Clinical implication:** Always explore for a second canal.



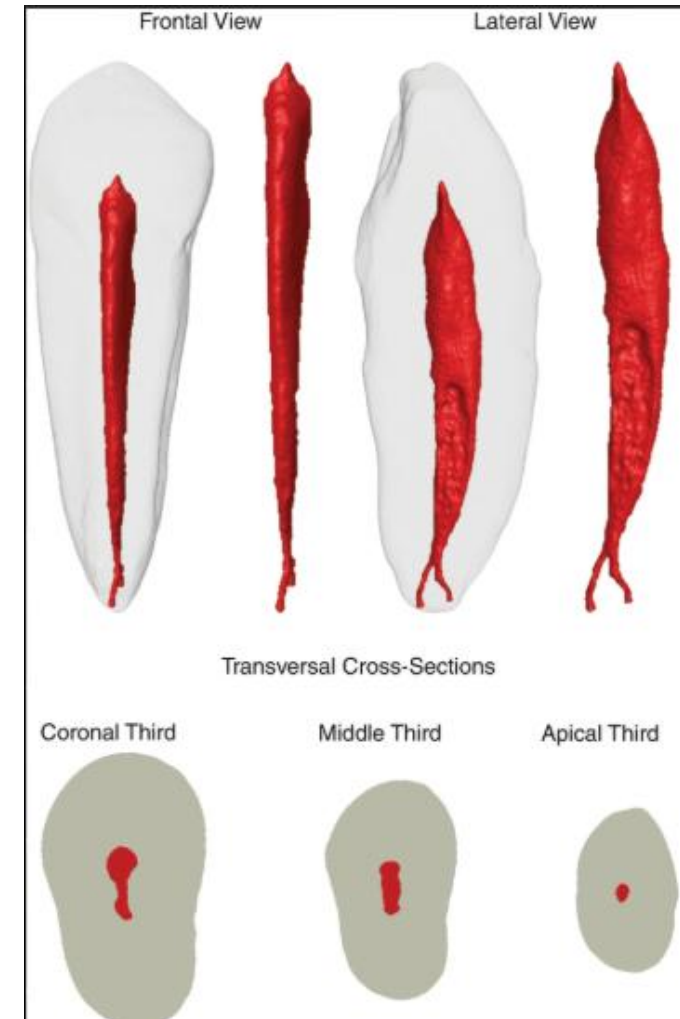
Mandibular Lateral Incisor – Pulp Morphology

- **Title:** Larger, Higher Incidence of Two Canals
- **Tooth and pulp:** Larger than mandibular central incisor.
- **Two canals:** More common – approximately **43%** .
- **Foramina:** When two canals are present, most exit by **separate foramina**.
- **Clinical implication:** Higher likelihood of two distinct canals with separate apical exits.



Mandibular Canine – Pulp Morphology

- **Title:** Similar to Maxillary Canine but Smaller
- **Pulp cavity:** Resembles maxillary canine but **smaller in all dimensions**.
- **Cross-section:** Oval, wider buccopalatally; becomes round apically.
- **Two canals:** Present in about **6%** .
 - Usually with **separate foramina**.
- **Clinical note:** Although less common, two canals should be considered.



References

1. Nelson SJ. The Maxillary Incisors. In: Nelson SJ, editor. Wheeler's Dental Anatomy, Physiology and Occlusion. 11th ed. St. Louis: Elsevier Saunders; 2020.
2. Berkovitz BKB. Maxillary Incisors. In: Berkovitz BKB, Holland GR, Moxham BJ, editors. Oral Anatomy, Histology and Embryology. 6th ed. Edinburgh: Mosby; 2025.

